Supplementary results: affine companions, spreads and exact-order statistics

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Abstract

This supplement accompanies *Norm-fiber densities and relative traces in characteristic two*. It provides coordinate Singer constructions, oriented-spread classifications of fixed complete mappings, a nonlinear companion pair generating A_{16}, and further scalar exact-order statistics. A separate trace-one geometric argument gives an alternative existence proof. The verification appendix also records the complete primitive inner-norm support theorem of the main paper and its separate certificates. A further appendix proves the new inner-norm moment and cubic-sieve distribution results.

Reading guide. Sections S1–S2 give the coordinate Singer constructions; Sections S3–S4 treat fixed mappings, oriented spreads and nonlinear companions. Sections S5–S7 collect the scalar consequences, single full-order statistic and alternative trace-one proof. Appendix S.A supplies the classical prescribed-trace existence input. The remaining appendices contain the extended field certificates (S.B), predecessor comparisons (S.C) and detailed verification record (S.D). Numbers with an S prefix refer to this supplement; unprefixed references refer to the main paper. Appendix S.E gives the complete proofs and exact spectrum certificate for the new inner-norm distributions.

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S1 The six-dimensional coordinate example

For the six-bit coordinate example, $V = \mathbb{F}_2^6$; other dimensions are stated explicitly. Coordinates $x = (x_1, \ldots, x_6)$ have x_1 the most significant bit (value 32) and x_6 the least (value 1); a word is identified with its integer value in $\{0, \ldots, 63\}$. Basis $G = (\mathtt{63}, \mathtt{33}, \mathtt{18}, \mathtt{49}, \mathtt{1}, \mathtt{21})$ (a fixed ordered basis — the six words are linearly independent over $\mathbb{F}_2$, by Gaussian elimination; non-uniqueness is discussed in §S2); write g_j for the j-th entry, $j = 0, \ldots, 5$. $\mathrm{coords}_G(x) = (c_1, \ldots, c_6)$ is the unique vector with $x = \bigoplus_{i=1}^{6} c_i\, g_{i-1}$, read as the six-bit word $c_1 \cdots c_6$ (c_1 most significant); $\oplus$ is bitwise XOR, i.e. addition in $\mathbb{F}_2^n$. When a field structure is chosen in characteristic two, $\oplus$ is also its field addition. A monic polynomial is called *primitive* if it is irreducible and its roots generate the multiplicative group of its root field. Superscripts on functions and on F denote composition iterates; orders of field elements denote multiplicative orders.

S1.1 An affine Singer map

Definition and basic checks

$T(x) = 63 \oplus \mathrm{coords}_G(x)$. Verified identities: $T(0) = 63$, $T(41) = 0$, $T(38) = 38$; for example $\mathrm{coords}_G(38) = (0, 1, 1, 0, 0, 1)$, read as $011001 = 25$, so $T(38) = 63 \oplus 25 = 38$, and $\mathrm{coords}_G(41) = (1, 1, 1, 1, 1, 1) = 63$, so $T(41) = 63 \oplus 63 = 0$. The integer labels $38, 41, 63$ belong to this chosen coordinate presentation; they are not invariants under changes of basis (see §S2).

Lemma S1.1. *T is affine: $T(x) = L_T x + b$ with $b = T(0) = 63$ and $L_T(x) = T(x) \oplus 63$.*

Proof. Since G is a basis, coords_G is an $\mathbb{F}_2$-linear isomorphism. Thus $L_T = \mathrm{coords}_G$, and adding the constant 63 gives the stated affine map. □

Lemma S1.2. $\mathrm{Fix}(T) = \{38\}$*; on the remaining* 63 *points T is a single cycle (type* $1 + 63$*, order* 63*), equivalently L_T has order* 63 *(Proposition S1.3).*

Proposition S1.3. *Identify $(V, +)$ with $\mathbb{F}_{64}$; then L_T is multiplication by a primitive element of $\mathbb{F}_{64}$, a Singer cycle [3]. The characteristic polynomial of L_T is*

$$p_T(t) = t^6 + t^5 + t^3 + t^2 + 1, \tag{S1.1}$$

computed directly as $\det(tI + L_T)$ *(code appendix), primitive of degree* 6*.*

Proof. The matrix in Appendix S.D gives the displayed determinant. Modulo p_T, repeated squaring and multiplication give

$$t^{64} = t, \qquad \gcd(p_T, t^8 + t) = \gcd(p_T, t^4 + t) = 1.$$

The degree-6 irreducibility criterion, using the prime divisors $2, 3$ of 6, therefore shows that p_T is irreducible. For completeness, $p_T(0) = p_T(1) = 1$ also gives $\gcd(p_T, t^2 + t) = 1$. In $\mathbb{F}_2[t]/(p_T)$ the residue $\bar{t}$ satisfies

$$\bar{t}^{63} = 1, \qquad \bar{t}^{21} = \bar{t}^4 + \bar{t}^2 + \bar{t} + 1 \neq 1, \qquad \bar{t}^9 = \bar{t}^2 + \bar{t} \neq 1.$$

Since $63 = 3^2 \cdot 7$, its order is 63. The irreducible characteristic polynomial identifies L_T with multiplication by $\bar{t}$ on this field. Finally, $T(38) = 38$ implies $T(x) \oplus 38 = L_T(x \oplus 38)$, proving the fixed-point and cycle claims of Lemma S1.2. □

Powers and $63 = 7 \cdot 9 = 3 \cdot 21$

Since L_T has order 63, T^d for $d \mid 63$ has type $1 + d \times (63/d)$ (one fixed point plus d cycles of length $63/d$): $T^1 = 1 + 63$, $T^3 = 1 + 3 \cdot 21$, $T^7 = 1 + 7 \cdot 9$, $T^9 = 1 + 9 \cdot 7$, $T^{21} = 1 + 21 \cdot 3$ (verified).

S1.2 The complete-mapping triple

A permutation f of an additive group of exponent two is a *complete mapping* if $\mathrm{id} \oplus f$ is again a permutation. Define its *triality operator* $F(f) = (\mathrm{id} \oplus f)^{-1}$, where $(\mathrm{id} \oplus f)(x) = x \oplus f(x)$. By Proposition S1.3, identify the linear part of T with multiplication by a primitive $\alpha \neq 1$. Then $\mathrm{id} \oplus T$ has nonzero slope $1 \oplus \alpha$ and is therefore a permutation. Its inverse $D = F(T)$ has slope $(1 \oplus \alpha)^{-1}$, and $1 \oplus (1 \oplus \alpha)^{-1} = \alpha/(1 \oplus \alpha) \neq 0$, so $\mathrm{id} \oplus D$ is also a permutation. Thus $D = F(T) = (\mathrm{id} \oplus T)^{-1}$ and $K = F(D) = (\mathrm{id} \oplus D)^{-1}$ are defined; since Lemma 2.2 gives $F^3(T) = T$, $\{T, D, K\}$ is the F-orbit of T. Verified facts:

- cycle types $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$ (seven 9-cycles plus one fixed point);
- fixed points $\mathrm{Fix}(T) = \{38\}$, $\mathrm{Fix}(D) = \{41\}$, $\mathrm{Fix}(K) = \{63\}$;
- $F(K) = T$, i.e. $(\mathrm{id} \oplus K)^{-1} = T$; hence $F^3 = \mathrm{id}$ with $T \to D \to K \to T$;
- offset-fixed three-cycle: $D(0) = 38 = \mathrm{Fix}(T)$, $K(0) = 41 = \mathrm{Fix}(D)$, $T(0) = 63 = \mathrm{Fix}(K)$;
- linear parts: for any $n \times n$ matrix A in characteristic two with $p_A(t) = \det(tI + A)$ and $p_A(1) \neq 0$, taking determinants gives

$$p_{(I+A)^{-1}}(t) = \frac{\det\big(t(I + A) + I\big)}{\det(I + A)} = \frac{t^n p_A(1 + t^{-1})}{p_A(1)}.$$

  Here $p_T(1) = 1$ and $p_T(t + 1) = p_T(t)$ (directly, $(t + 1)^6 + (t + 1)^5 + (t + 1)^3 + (t + 1)^2 + 1$ reduces to $t^6 + t^5 + t^3 + t^2 + 1$). Hence $L_D = (I + L_T)^{-1}$ has $p_D(t) = t^6 p_T(1 + t^{-1}) = t^6 p_T(t^{-1}) = t^6 + t^4 + t^3 + t + 1$ (primitive). The reciprocal form uses this shift invariance. L_K has $\Phi_9(t) = t^6 + t^3 + 1$ (cyclotomic, non-primitive, order 9).

For the Singer example, in field coordinates $T(x) = \alpha x \oplus c$, so $F(T)(0) = D(0) = (1 \oplus \alpha)^{-1} c$ solves $x = \alpha x \oplus c$, i.e. $D(0) = \mathrm{Fix}(T)$, and the slope cycle $\alpha \mapsto (1 \oplus \alpha)^{-1} \mapsto 1 \oplus \alpha^{-1} \mapsto \alpha$ gives the two remaining identities. For the six-bit layer the values are $F(T)(0) = 38$, $F(D)(0) = 41$, $F(K)(0) = 63$. Conceptually, T is the launching map — an affine Singer layer read through the fixed generator basis (a change of basis) — while the triality operator F keeps the third-order cycle running: $T \mapsto D \mapsto K \mapsto T$, carrying the fixed point $38 \mapsto 41 \mapsto 63 \mapsto 38$.

Theorem S1.4 (The six-dimensional Singer application)**.** $\langle T, D \rangle = \mathrm{AGL}(1, 64)$*; its order is* $4032 = 64 \cdot 63$.

Proof. By Proposition S1.3, in field coordinates the linear part of T is multiplication by a primitive $\alpha \in \mathbb{F}_{64}$. The translation is $c = T(0) = 63 \neq 0$. Thus $U = V$ and the least common multiple in Corollary 2.4 is already 63 because $\mathrm{ord}(\alpha) = 63$. The corollary proves the claim. □

The general structure allows reducible and nonsemisimple matrices.

The exponent convention

For the six-bit example, the fixed point 38 can serve as the origin after relabeling, while 0 remains the additive origin. In the value set $\{0, \dots, 63\}$ the 9-multiples are exactly the 8 self-doubled words $(x_i = x_{i+3})$, a 3-dimensional subspace. To parametrize the long orbit, define

$$\theta(j) = T^j(63), \qquad j \in \mathbb{Z}/63.$$

This is a bijection onto $V \setminus \{38\}$. Apply the CRT isomorphism to the *iteration exponent* j, sending it to $(j \bmod 7, j \bmod 9)$. Thus $\theta(0) = 63$ corresponds to $(0, 0)$, whereas $0 = \theta(62)$ corresponds to $(6, 8)$, since $T(0) = 63$. Reduction of the integer word labels modulo 63 does not give this parametrization. The cyclotomic factor Φ_{2^k+1} reflects $(2^k - 1)(2^k + 1) = 2^{2k} - 1$; the fixed point of K accounts for the remaining point.

S2 Uniform statement for $2k$-bit words

Identify the $2k$-dimensional $\mathbb{F}_2$-space V_{2k} with $\mathbb{F}_{2^{2k}}$; write $M = 2^{2k} - 1 = (2^k - 1)(2^k + 1)$. In field coordinates an affine Singer layer is a map $T(x) = \alpha x \oplus b$ whose linear part is multiplication by a primitive element α (order M).

Lemma S2.1. *For every $k \geq 1$, $\mathrm{ord}_{2^k+1}(2) = 2k$.*

Proof. Let $m = \mathrm{ord}_{2^k+1}(2)$. Since $2^k \equiv -1 \pmod{2^k + 1}$, we have $m \mid 2k$, while $m \nmid k$. Suppose, for contradiction, that $m < 2k$. If m were odd, $m \mid 2k$ would imply $m \mid k$. Thus m is even; write $m = 2d$ with $d \mid k$, and $m < 2k$ gives $d < k$. The subgroup $\langle 2 \rangle \subset (\mathbb{Z}/(2^k+1)\mathbb{Z})^\times$ is cyclic of order $2d$, so its unique element of order 2 is 2^d. But $2^k \equiv -1$ also has order 2 in this subgroup; hence $2^d \equiv -1 \pmod{2^k + 1}$. Thus $2^k + 1 \mid 2^d + 1$, contradicting $0 < 2^d + 1 < 2^k + 1$. Therefore $m = 2k$. □

For $q = 2^k$, define

$$\mathrm{PTE}(k): \quad \exists \beta \in \mathbb{F}_{q^2}^*, \quad \mathrm{ord}(\beta) = q^2 - 1, \quad \mathrm{Tr}_{\mathbb{F}_{q^2}/\mathbb{F}_q}(\beta) = 1.$$

The following theorem supplies the required existence input; its proof is given in Appendix S.A.

Theorem S2.2 (Classical PTE existence)**.** *For every $k \geq 1$, $\mathrm{PTE}(k)$ holds.*

Proof. See Appendix S.A. □

Theorem S2.3. *For every $k \geq 1$ there exists a primitive $\alpha \in \mathbb{F}_{2^{2k}}$ such that, for every $b \in \mathbb{F}_{2^{2k}}^*$, the affine Singer layer $T_b(x) = \alpha x \oplus b$ has $D_b = (\mathrm{id} \oplus T_b)^{-1}$ and $K_b = (\mathrm{id} \oplus D_b)^{-1}$ defined. The maps T_b, D_b have cycle type $1 + M$, while K_b has cycle type $1 + (2^k - 1) \times (2^k + 1)$, and $\langle T_b, D_b \rangle = \mathrm{AGL}(1, 2^{2k})$. The triality operator satisfies $F^3 = \mathrm{id}$ with $T_b \mapsto D_b \mapsto K_b \mapsto T_b$.*

Proof. Let $b \neq 0$ be arbitrary and write $T = T_b$, $D = D_b$, $K = K_b$. The linear parts of T, D, K are α, $(1 \oplus \alpha)^{-1}$, and $\kappa = 1 \oplus \alpha^{-1} = (1 \oplus \alpha)/\alpha$ respectively (independent of the translation b); the three slopes are cycled by the map $z \mapsto 1/(z \oplus 1)$, so $F^3 = \mathrm{id}$. Since $\mathbb{F}_{2^{2k}}^*$ is cyclic, an affine map with slope $\mu \neq 1$ has cycle type $1 + \frac{M}{\mathrm{ord}(\mu)} \times \mathrm{ord}(\mu)$ (the translation only moves the fixed point; a shorter orbit would force T^d, $d < \mathrm{ord}(\mu)$, to have a second fixed point, impossible since $\mu^d \neq 1$); the stated types therefore follow once $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = M$ and $\mathrm{ord}(\kappa) = 2^k + 1$. Existence of such an α follows directly from Theorem S2.2. Put $q = 2^k$ and choose a primitive $\beta \in \mathbb{F}_{q^2}$ with $\beta^q \oplus \beta = 1$. Take $\alpha = \beta$. Then $1 \oplus \alpha = \alpha^q$, so

$$\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = q^2 - 1 = M, \qquad \kappa = \frac{1 \oplus \alpha}{\alpha} = \alpha^{q-1}, \qquad \mathrm{ord}(\kappa) = q + 1.$$

Here the last equality follows from $\mathrm{ord}(\alpha^{q-1}) = M/\gcd(M, q-1) = q+1$. These are exactly the required slope orders. To make the dependence on b explicit, set $S_b(x) = bx$. Then $T_b = S_b T_1 S_b^{-1}$. Indeed, for any invertible additive map S and $g = SfS^{-1}$,

$$(\mathrm{id} \oplus g)(x) = S\big((\mathrm{id} \oplus f)(S^{-1}x)\big).$$

Taking inverses gives $F(g) = SF(f)S^{-1}$ whenever $F(f)$ is defined. Applying this with $S = S_b$ gives $D_b = S_b D_1 S_b^{-1}$ and $K_b = S_b K_1 S_b^{-1}$. Their fixed points are

$$\mathrm{Fix}(T_b) = \{b/(1 \oplus \alpha)\}, \qquad \mathrm{Fix}(D_b) = \{b/\alpha\}, \qquad \mathrm{Fix}(K_b) = \{b\}.$$

Thus all nonzero b give the stated cycle types; §2 gives the group-generation claim. □

Since the K-slope has order 2^k+1, the degree over $\mathbb{F}_2$ of its minimal polynomial is $\mathrm{ord}_{2^k+1}(2) = 2k$ by Lemma S2.1; this polynomial is therefore an irreducible degree-$2k$ factor of Φ_{2^k+1}. **Remark (the τ-collapse).** Put $q = 2^k$. For $\gamma \in \mathbb{F}_{q^2}^*$ of order $q+1$, one has $\gamma^q = \gamma^{-1}$; hence $\tau = \gamma \oplus \gamma^{-1}$ satisfies $\tau^q = \tau$ and belongs to $\mathbb{F}_q$. The Frobenius identities $(1 \oplus \gamma)^{2^k-1} = \gamma^{-1}$ and $(1 \oplus \gamma)^{2^k+1} = \tau$, together with $\gcd(2^k-1, 2^k+1) = 1$, give the following order calculation. For an element z of finite order N and coprime positive integers r, s, $\mathrm{ord}(z^r) = N/\gcd(N, r)$ implies $N = \mathrm{lcm}(\mathrm{ord}(z^r), \mathrm{ord}(z^s))$. Consequently $\mathrm{ord}(1 \oplus \gamma) = \mathrm{lcm}(\mathrm{ord}(\gamma^{-1}), \mathrm{ord}(\tau)) = \mathrm{lcm}(2^k+1, 2^k-1) = M$ whenever τ is primitive in $\mathbb{F}_{2^k}$; the pair $(u^{2^k-1}, u^{2^k+1}) = (\gamma^{-1}, \tau^{-1})$ for $u = \gamma/(1 \oplus \gamma)$ gives the same order for u. Since $\alpha = (1 \oplus \gamma)^{-1}$ and $\kappa = 1 \oplus \alpha^{-1} = \gamma$, once γ has order 2^k+1 the three order conditions on α reduce to the single condition that τ be primitive. The order assumption on γ is essential; it is not implied by τ being primitive alone. Existence of such a γ is the classical square-order statement (see [19] and [20, Theorem 1, p. 1]; for an equivalent square-order formulation, see [18, Theorem 1.5, p. 170]); it is also supplied directly by Theorem S2.2. For $q = 2^k$ the appendix proof is self-contained, so the paper does not use these classical results as external existence inputs.

For this coordinate application, an ordered generator basis is called *admissible for the stated Singer triple* when the resulting linear slopes satisfy $\mathrm{ord}(\alpha) = \mathrm{ord}(1+\alpha) = 2^{2k}-1$ and $\mathrm{ord}(1+\alpha^{-1}) = 2^k+1$. These order conditions specify exactly which coordinate presentations realize the theorem's triple. The argument of §2 applies verbatim with $Q = 2^{2k}$ (taking $b \neq 0$ in the proof), so the generation $\langle T, D\rangle = \mathrm{AGL}(1, 2^{2k})$ holds for every k. For the concrete coordinate presentation of §S1.1 the orders and cycle data were verified for $k = 1, 2, 3, 4$ (the bases are listed in the appendix): $k = 1$ (2 bits) order $12 = 4 \cdot 3$, K type $1+1 \cdot 3$, factor Φ_3 (basis $(1, 3)$: $\mathrm{Fix}(T) = \{1\}$, $\mathrm{Fix}(D) = \{2\}$, $\mathrm{Fix}(K) = \{3\}$); $k = 2$ (4 bits) order $240 = 16 \cdot 15$, K type $1 + 3 \cdot 5$, factor Φ_5; $k = 3$ (6 bits) order $4032 = 64 \cdot 63$, K type $1 + 7 \cdot 9$, factor Φ_9 (this paper); $k = 4$ (8 bits) order $65280 = 256 \cdot 255$, K type $1 + 15 \cdot 17$, linear part an irreducible degree-8 factor of Φ_{17}.

The basis is not canonical even in dimension 6: among the 720 orderings of these six generators, 572 do not even make $\mathrm{id} \oplus T$ a permutation; of the 148 that do, only 54 give T of order 63, and only 9 give the exact cycle-type triple $T = D = 1 + 63$, $K = 1 + 7 \cdot 9$. For instance $(4\,6)$ keeps $T = 1 + 63$ but changes $\mathrm{Fix}(T)$ from 38 to 11 and the cycle types of D and K to $D = 1 + 3 \cdot 21$, $K = 1 + 63$. Hence the particular entries of G and the anchors $\{38, 41, 63\}$ are coordinates of one representative, not invariants, and no uniqueness statement is made. The identity $F^3 = \mathrm{id}$ holds for every complete mapping. The group order and the specified cycle triple follow for the admissible presentations defined above; completeness alone does not impose those three slope orders. The general formula of Theorem 2.3 applies to every presentation, and determines its actual group.

Remark (constructiveness).
Theorem S2.3 is an existence statement; its existence input is proved in Appendix S.A. The τ-collapse turns it into a finite search with certification. For $\tau = \gamma \oplus \gamma^{-1} \in \mathbb{F}_{2^k}$, the Artin–Schreier identity

($x^2 + \tau x + 1$ *is irreducible over* $\mathbb{F}_{2^k}$ *iff* $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(1/\tau) = 1$) gives 2^{k-1} candidate values of τ. For a candidate, fix an explicit representation of $\mathbb{F}_{2^{2k}}$ and take a root γ of $x^2 + \tau x + 1$ (found, for instance, by exhaustive search); the collapse gives the required orders for $1 \oplus \gamma$ and $\gamma/(1 \oplus \gamma)$ precisely when τ is primitive *and* $\mathrm{ord}(\gamma) = 2^k + 1$. The second condition is not automatic from the first: for example, when $k = 6$ and $\mathbb{F}_{64} = \mathbb{F}_2[t]/(t^6 + t + 1)$, the element $\tau = t^5 + t^3 + 1$ is primitive and $\mathrm{Tr}(1/\tau) = 1$, but the roots of $x^2 + \tau x + 1$ have order 13, not 65. The theorem of Appendix S.A guarantees that a jointly successful candidate exists, so a deterministic enumeration that tests both conditions terminates. Certifying them requires factoring $2^k - 1$ and $2^k + 1$ (or an equivalent exact-order test). This procedure supplies a deterministic construction with certification; no factorization-free closed formula is supplied here. A necessary restatement is that $u = 1/\tau$ is a primitive element of $\mathbb{F}_{2^k}$ with $\mathrm{Tr}_{\mathbb{F}_{2^k}/\mathbb{F}_2}(u) = 1$; equivalently, its degree-k minimal polynomial has x^{k-1}-coefficient 1. This condition alone is still not sufficient for admissibility, because the root-order condition above must also be checked.

S3 Fixed mappings and orbits under F

Lemma 2.2 implies that every orbit under F has length one or three. We first give the scalar classification, then describe all fixed complete mappings geometrically and count the orbits in the entire affine class. An *F-fixed map* means $F(f) = f$; this is distinct from the unique fixed point of the permutation f in Lemma 2.1.

S3.1 Coordinate-permutation symmetry

Proposition S3.1. *For a complete mapping f of an additive group V of exponent two, put*

$$\Gamma_f = \{(x, f(x), x \oplus f(x)) : x \in V\}, \qquad J(f) = f^{-1}.$$

Permuting the three coordinates preserves this family of sets. The permutations $\rho(x, y, z) = (z, x, y)$ and $\sigma(x, y, z) = (y, x, z)$ induce F and J, respectively. Thus J preserves complete mappings and

$$F^3 = J^2 = \mathrm{id}, \qquad JFJ = F^{-1} = F^2.$$

For finite V, Γ_f is a transversal of its additive Cayley table.

Proof. Every triple of Γ_f has sum zero, and each of its three coordinate projections is a bijection to V. Both properties are invariant under coordinate permutations. Conversely these properties recover a unique complete mapping from the first two coordinates. With $z = x \oplus f(x)$, one has $F(f)(z) = x$ and $z \oplus x = f(x)$, so

$$\rho(\Gamma_f) = \Gamma_{F(f)}.$$

With $y = f(x)$, swapping the first two coordinates gives $(y, f^{-1}(y), y \oplus f^{-1}(y))$, hence $\sigma(\Gamma_f) = \Gamma_{J(f)}$. The coordinate permutations satisfy $\rho^3 = \sigma^2 = \mathrm{id}$ and $\sigma\rho\sigma = \rho^{-1}$; injectivity of $f \mapsto \Gamma_f$ transfers these relations to F, J. For finite V, the three bijective projections say exactly that the set contains one cell in each row, each column, and each symbol class of the additive Cayley table. This is the definition of a transversal. □

This is the cyclic part of the classical coordinate-permutation symmetry of Latin squares [1], acting here on transversals of a table invariant under all six coordinate permutations. The proof identifies this relationship explicitly; the order-three symmetry itself is not claimed as new.

Scalar affine maps

Proposition S3.2. *Let $k \geq 1$, $Q = 2^{2k}$, and $\mathcal{A}_Q = \{f_{a,c} : x \mapsto ax \oplus c \mid a \in \mathbb{F}_Q \setminus \{0,1\},\ c \in \mathbb{F}_Q\}$. The operator F acts on $\mathcal{A}_Q$ with two singleton orbits and $(Q(Q-2)-2)/3$ orbits of length three. Its fixed maps are exactly $f_{a,0}$ with $a^2+a+1=0$. On the invariant subset $c \neq 0$, every orbit has length three.*

Proof. In coefficient pairs,

$$F(a,c) = \left((1+a)^{-1}, \frac{c}{1+a}\right).$$

The new slope again lies outside $\{0,1\}$, and Lemma 2.2 gives $F^3 = \mathrm{id}$, so each orbit has length one or three. A fixed pair must satisfy $a = (1+a)^{-1}$, equivalently $a^2+a+1=0$; the constant-term equation is $c(1+a) = c$, so $ac = 0$ and $c = 0$. Conversely these conditions give a fixed pair. The polynomial a^2+a+1 has two distinct roots in $\mathbb{F}_Q$, since $\mathbb{F}_4 \subseteq \mathbb{F}_Q$. There are $Q(Q-2)$ pairs in total, yielding the stated counts. Multiplication of c by $(1+a)^{-1} \neq 0$ preserves $c \neq 0$, and no fixed pair lies in that subset. □

Fixed maps and line spreads

A *line spread* of $V = \mathbb{F}_2^n$ is a collection of two-dimensional $\mathbb{F}_2$-subspaces whose nonzero vectors partition $V \setminus \{0\}$. Thus a line here is two-dimensional as a vector space; it is a projective line in $\mathrm{PG}(n-1,2)$. Each such subspace has exactly three nonzero vectors $x, y, x \oplus y$. A cyclic orientation on a line specifies one of the two 3-cycles on them.

Proposition S3.3. *For $n \geq 1$, the F-fixed complete mappings of $\mathbb{F}_2^n$ are in bijection with line spreads equipped with a cyclic orientation on each line. Every such map fixes 0 and has only 3-cycles on the nonzero vectors. They exist exactly when n is even. If s_n denotes the number of line spreads in a fixed coordinate space, their number, for even n, is*

$$s_n \, 2^{(2^n-1)/3}.$$

Nonlinear F-fixed complete mappings exist for every even $n \geq 4$.

Proof. If $F(f) = f$, then $f^{-1} = \mathrm{id} \oplus f$. Evaluating at $f(x)$ gives

$$f^2(x) = x \oplus f(x) = f^{-1}(x), \qquad f^3 = \mathrm{id}.$$

Let p be the unique fixed point from Lemma 2.1. The displayed equality at p gives $p = p \oplus p = 0$. Thus each nonzero cycle is $(x, f(x), x \oplus f(x))$, the nonzero part of a two-dimensional subspace. Disjoint cycles supply a line spread and their orientations. Conversely, orient each line of a spread and fix 0. The resulting permutation satisfies $f^2 = \mathrm{id} \oplus f = f^{-1}$, so it is complete and $F(f) = f$. This proves the bijection and the count.

A spread requires $3 \mid 2^n - 1$, hence even n. If $n = 2m$, identify V with $\mathbb{F}_4^m$; its one-dimensional $\mathbb{F}_4$-subspaces give a line spread over $\mathbb{F}_2$, proving existence. For $m \geq 2$ (equivalently, even $n \geq 4$), start with the linear map $A : x \mapsto \omega x$, where $\omega^2 + \omega + 1 = 0$ in $\mathbb{F}_4$, and reverse the orientation on exactly one spread line U. The resulting map f equals A^2 on $U \setminus \{0\}$ and equals A elsewhere, and remains F-fixed. For $0 \neq u \in U$ and $v \notin U$, one has $u \oplus v \notin U$, and

$$f(u \oplus v) = Au \oplus Av \neq A^2 u \oplus Av = f(u) \oplus f(v),$$

since $(A^2 + A)u = u \neq 0$. Hence f is nonlinear. □

If V is identified with $\mathbb{F}_{2^n}$ for even n, the spread obtained from $\mathbb{F}_4$ consists of the multiplicative cosets of $\mathbb{F}_4^*$. An orientation on each coset is multiplication by either ω or ω^2. Thus these examples are a particular family of the first-order cyclotomic complete mappings considered in [22]. Proposition S3.3 describes fixed maps from arbitrary line spreads, without choosing a field structure or restricting to this coset partition.

For $n = 4$, exact-cover enumeration of the 35 lines on 15 nonzero vectors gives $s_4 = 56$, recovering the classical spread count also formalized in Coq by Magaud [2]. The proposition then gives $56 \cdot 2^5 = 1792$ fixed complete mappings. Enumeration of their orientations finds 112 linear and 1680 nonlinear maps, in agreement with the affine count below. These finite counts, the complete list of spreads, and a nonlinear witness are reproduced by `verify_complete_mapping_orbits.py`.

All affine complete mappings

Write $h_j = |\mathrm{GL}(j,2)| = \prod_{i=0}^{j-1}(2^j - 2^i)$, where $h_0 = 1$ is the empty product, and let

$$\mathcal{C}_n^{\mathrm{aff}} = \{x \mapsto Lx \oplus c : \ L, I + L \in \mathrm{GL}(n,2),\ c \in \mathbb{F}_2^n\}.$$

This includes matrices that are not scalar multiplication in a single field structure on $\mathbb{F}_2^n$.

Proposition S3.4. *For $n \geq 1$, put*

$$d_n = h_n \sum_{j=0}^{n} \frac{(-1)^j 2^{j(j-1)/2}}{h_j}, \qquad r_n = \begin{cases} 0, & n \text{ odd}, \\ h_n/|\mathrm{GL}(n/2,4)|, & n \text{ even}. \end{cases}$$

There are $2^n d_n$ affine complete mappings. Under F they form r_n singleton orbits and $(2^n d_n - r_n)/3$ orbits of length three. The fixed maps are exactly $x \mapsto Lx$ with $L^2 + L + I = 0$.

Proof. The number d_n counts invertible matrices with no nonzero fixed vector. Its displayed expression is the zero-dimensional fixed-space case of the classical Rudvalis–Shinoda count, recorded in [24, Theorem 2.1]. Here is a direct proof. For a j-dimensional subspace U put $\mu(U) = (-1)^j 2^{j(j-1)/2}$. If W has dimension r, the Gaussian-binomial identity gives

$$\sum_{U \leq W} \mu(U) = \sum_{j=0}^{r} \binom{r}{j}_2 (-1)^j 2^{j(j-1)/2} = \prod_{i=0}^{r-1}(1 - 2^i).$$

This is 1 when $W = 0$ and 0 otherwise. Apply it to $W = \ker(I + L)$ and sum over $L \in \mathrm{GL}(n,2)$. For a fixed j-subspace U, the matrices fixing U pointwise have block form $\left(\begin{smallmatrix} I & B \\ 0 & C \end{smallmatrix}\right)$, with $2^{j(n-j)}$ choices of B and h_{n-j} choices of C. Since there are $\binom{n}{j}_2$ choices of U and

$$\binom{n}{j}_2 2^{j(n-j)} h_{n-j} = h_n / h_j,$$

interchanging the two finite sums yields the formula for d_n. Each such L permits all 2^n translations c.

For $f(x) = Lx \oplus c$, the condition $F(f) = f$ is equivalent to $(\mathrm{id} \oplus f) \circ f = \mathrm{id}$. Comparing linear and constant parts gives $L^2 + L = I$ and $Lc = 0$, so $c = 0$. The polynomial $z^2 + z + 1$ is irreducible over $\mathbb{F}_2$. Thus such an L equips V with an $\mathbb{F}_4$-vector-space structure, which requires even n. Conversely, for even n multiplication by a root of this polynomial supplies such an L. All these matrices form one conjugacy class in $\mathrm{GL}(n,2)$: choose an $\mathbb{F}_4$-basis in each structure and compare the resulting $\mathbb{F}_2$-bases. The invertible maps commuting with L are exactly the $\mathbb{F}_4$-linear ones, so the centralizer has order $|\mathrm{GL}(n/2,4)|$. This proves the formula for r_n. Finally Lemma 2.2 restricts the other orbits to length three. □

The following small values are exact evaluations of the formulas, independently checked by enumerating all binary matrices and translations in these dimensions:

n	d_n	$\lvert\mathcal{C}_n^{\mathrm{aff}}\rvert$	F-fixed maps	3-element orbits
2	2	8	2	2
3	48	384	0	128
4	5824	93184	112	31024

Every F-fixed affine map is linear. In particular the 1680 nonlinear fixed maps in dimension four lie outside the affine class, showing the additional scope of Proposition S3.3.

Linear conjugacy of oriented spreads

We now distinguish the F-orbits of maps from equivalence under a linear change of coordinates. Write $f \sim g$ if $g = SfS^{-1}$ for some $S \in \mathrm{GL}(n,2)$. Such conjugacy preserves $F(f) = f$. For a spread $\mathcal{S}$, let $G_{\mathcal{S}}$ be its setwise stabilizer in $\mathrm{GL}(n,2)$, and choose a reference cyclic orientation on each line. An orientation is then a vector $(\epsilon_L)_{L\in\mathcal{S}}$ over $\mathbb{F}_2$, with 1 meaning reversal of the reference orientation. For $g \in G_{\mathcal{S}}$ define $\delta_g(L) \in \mathbb{F}_2$ by comparing the transported reference orientation on gL with the chosen reference on gL. Thus the transported bits satisfy

$$\epsilon'_{gL} = \epsilon_L \oplus \delta_g(L).$$

Proposition S3.5 (Orientation-sensitive orbit formula)**.** *The correspondence of Proposition S3.3 identifies linear conjugacy with equivalence of oriented spreads. Let $c(g)$ be the number of cycles in the permutation of spread lines induced by g, and put*

$$e(g) = \begin{cases} 1, & \bigoplus_{L\in C} \delta_g(L) = 0 \text{ for every line cycle } C, \\ 0, & \text{otherwise.} \end{cases}$$

The number of linear-conjugacy classes supported on the orbit of $\mathcal{S}$ is

$$a(\mathcal{S}) = \frac{1}{|G_{\mathcal{S}}|} \sum_{g\in G_{\mathcal{S}}} e(g) 2^{c(g)}.$$

Summing this over representatives of the linear equivalence classes of underlying spreads gives all F-fixed linear-conjugacy classes. The stabilizer of a fixed map is the stabilizer of its oriented spread.

Proof. Linear conjugacy sends each cycle $(x, f(x), x \oplus f(x))$ to its transported cycle, so it sends the spread and its orientations to exactly those of SfS^{-1}. To fix an orientation, the displayed bit relation must hold with $\epsilon' = \epsilon$. Around a line cycle the successive equations are consistent if and only if the sum of the $\delta_g(L)$ on that cycle is zero. When consistent, there are two choices for its first bit and all other bits are forced. Thus g fixes $e(g)2^{c(g)}$ orientations. Burnside's orbit-counting lemma gives the formula. Restricting to the stabilizer of one underlying spread suffices for its full linear orbit; distinct underlying spread orbits cannot give conjugate maps. The same transport correspondence gives the stabilizer assertion. □

Proposition S3.6 (Local criterion for an invariant orientation)**.** *Let $\mathcal{S}$ be any line spread and $G = G_{\mathcal{S}}$ its setwise stabilizer in $\mathrm{GL}(V)$. For $L \in \mathcal{S}$, write $G_L = \{g \in G : gL = L\}$. The following conditions are equivalent:*

1. *There is an orientation of every line in $\mathcal{S}$ that is fixed by all of G.*

2. *Every $g \in G_L$, for every $L \in \mathcal{S}$, induces an even permutation of the three nonzero vectors of L.*

3. *The orientation-translation cocycle associated with any choice of reference orientations is a coboundary.*

More precisely, put $E = \mathbb{F}_2^{\mathcal{S}}$ with its natural permutation G-module structure $(g \cdot v)_{gL} = v_L$. If $D_g \in E$ is defined by $(D_g)_{gL} = \delta_g(L)$, then $D_{gh} = D_g + g \cdot D_h$, and condition (3) means that there exists $v \in E$ with $D_g = v + g \cdot v$ for every $g \in G$.

Proof. An orientation of a line is a cyclic ordering of its three nonzero vectors, up to cyclic rotation. Its stabilizer within the permutation group on those vectors is the alternating group of degree three. Hence (1) implies (2).

Assume (2). Choose one representative L from each G-orbit on $\mathcal{S}$ and choose an arbitrary cyclic orientation on it. Transport the chosen orientation to gL using $g \in G$. If $gL = hL$, then $h^{-1}g \in G_L$ and (2) says that this element preserves the chosen orientation. Thus the transport is independent of the choice of g, and the resulting orientation on all lines is fixed by G. This proves (2)⇒(1).

For the last equivalence, the transported bit vector is $g \cdot v + D_g$. Composing two transports gives $D_{gh} = D_g + g \cdot D_h$. The bit vector v is fixed by every g precisely when $v = g \cdot v + D_g$, or equivalently $D_g = v + g \cdot v$. This is exactly the coboundary condition. □

The proposition distinguishes the existence of one orientation fixed by the entire spread automorphism group from the individual fixed-orientation tests $e(g)$ in Proposition S3.5; the latter still supply the general orbit count even when no globally invariant orientation exists.

Theorem S3.7 (Four-dimensional classification)**.** *There are exactly three conjugacy classes under* $\mathrm{GL}(4,2)$ *of F-fixed complete mappings of $\mathbb{F}_2^4$. Identify V with $\mathbb{F}_4^2$, choose $\omega^2 + \omega + 1 = 0$, and start with $f_0(x) = \omega x$ on its five spread lines. Representatives are obtained by reversing any r lines, with $r = 0, 1, 2$. Their orbit sizes, stabilizer orders and stabilizer groups are*

r	*orbit size*	*stabilizer order*	*stabilizer group*	*linear*
0	*112*	*180*	$C_3 \times A_5$	*yes*
1	*560*	*36*	$C_3 \times A_4$	*no*
2	*1120*	*18*	$C_3 \times S_3$	*no*

Here C_3 is cyclic of order three, and A_j, S_j denote alternating and symmetric groups. In particular the 1680 *nonlinear maps form two classes.*

Proof. First every four-dimensional spread is equivalent to the displayed one. Choose two spread lines as complementary spaces $E \oplus E$, where $E = \mathbb{F}_2^2$. The other three lines are graphs of invertible maps $P_i : E \to E$, with all $P_i + P_j$ invertible. A change of coordinates makes one of these maps I. Among the six matrices of $\mathrm{GL}(2,2)$, the only ones for which $I + P$ is invertible are

$$P = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad P^2 = \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}.$$

They satisfy $P^2 + P + I = 0$. The three graphs are consequently those of I, P, P^2, giving the five one-dimensional $\mathbb{F}_4$-subspaces of $\mathbb{F}_4^2$.

For this standard spread, the only linear maps inducing cyclic orientations on its lines are f_0 and f_0^2. Indeed on each line such a map is multiplication by ω or ω^2; additivity at $e_1 + e_2$ forces the choices on the two coordinate lines to agree, which determines the map on their direct sum. Therefore any linear stabilizer of the underlying spread conjugates f_0 to either f_0 or f_0^2. The commuting

maps are $\mathrm{GL}(2,4)$; composing with the coordinatewise Frobenius map $\sigma(x_1,x_2)=(x_1^2,x_2^2)$ accounts for the other coset. Thus the full spread stabilizer is $\Gamma\mathrm{L}(2,4)$ of order $2|\mathrm{GL}(2,4)|=360$. Since $|\mathrm{GL}(4,2)|=20160$, there are $20160/360=56$ spreads, recovering the classical count [2].

On the five lines, $\mathrm{GL}(2,4)$ has scalar kernel C_3 and image $\mathrm{PGL}(2,4)$ of order 60. This index-two subgroup of S_5 is A_5. Its elements preserve every reference orientation. Frobenius reverses them all and induces the transposition exchanging ω and ω^2 on $\mathbb{P}^1(\mathbb{F}_4)$, fixing $0,1,\infty$. Consequently even line permutations permute the reversed-line subset, whereas odd line permutations also complement it. A_5 is transitive on subsets of each fixed size: an arbitrary permutation carrying one subset to another can have its parity adjusted by a transposition inside the subset or its complement. Hence the orientation orbits are exactly the weight pairs $\{0,5\}$, $\{1,4\}$, $\{2,3\}$, of sizes $2,10,20$. Multiplying by the 56 underlying spreads gives the three displayed orbit sizes and orbit–stabilizer gives their stabilizer orders.

For the representatives of weights $r=0,1,2$, an odd line permutation cannot stabilize the orientation because it changes the weight to $5-r$. Within $\mathrm{GL}(2,4)$ the projective stabilizers are A_5, A_4, and the two-subset stabilizer S_3, respectively. The determinant splits $\mathrm{GL}(2,4)=C_3\times\mathrm{SL}(2,4)$; the projective action identifies $\mathrm{SL}(2,4)$ with A_5, giving the displayed direct products. The earlier two-linear-map observation shows that only the first class is linear. □

The proof gives representatives and stabilizers, rather than only the total number of maps. Independently, `verify_affine_groups_and_conjugacy.py` constructs all 1792 maps and conjugates the three representatives by all 20160 matrices; their disjoint orbits exhaust the list. It also compares direct fixed-orientation counts with Proposition S3.5 for all 360 standard-spread automorphisms. For dimension six, underlying spreads have already been classified by Mateva and Topalova [25] into 131044 types. We next treat the standard Desarguesian type, one of these types. The other underlying types are outside the six-dimensional computation reported here.

Affine group orders in dimension four

group order	3	12	15	48	192	240
number of maps	112	3360	4032	5040	20160	60480

Of the 93184 affine complete mappings, 80640 give transitive groups; 60480 give primitive groups, all doubly transitive in this dimension. These counts are computational evaluations of the structure theorem, reproduced by the additional verifier in Appendix S.D; the action criteria are proved in main Corollary 2.4 for every dimension.

Standard spreads in arbitrary even dimension

Let $V=\mathbb{F}_4^m$, let $\omega^2+\omega+1=0$, and let $\mathcal{S}_m$ consist of its one-dimensional $\mathbb{F}_4$-subspaces, viewed as two-dimensional $\mathbb{F}_2$-subspaces. Put

$$N_m=\frac{4^m-1}{3},\qquad P_m=\mathrm{PGL}(m,4),\qquad p_m=|P_m|=\frac{1}{3}\prod_{i=0}^{m-1}(4^m-4^i).$$

Orient each line by $x\mapsto\omega x$ before choosing the reversed-line subset. Write a_m for the number of full linear-conjugacy classes supported on spreads equivalent to $\mathcal{S}_m$.

Theorem S3.8 (Standard-spread action and asymptotics)**.** *For $m\geq 2$, the setwise stabilizer of $\mathcal{S}_m$ in $\mathrm{GL}(2m,2)$ is $\Gamma\mathrm{L}(m,4)$. Its linear coset permutes orientation bits; its semilinear coset also complements every bit and fixes no orientation. Every oriented stabilizer contains the scalar group*

$C_3 = \mathbb{F}_4^*$ and is contained in $\mathrm{GL}(m,4)$. *If $c(g)$ counts the cycles of g on the N_m projective points, then*

$$a_m = \frac{1}{2p_m} \sum_{g \in P_m} 2^{c(g)} = \frac{2^{N_m}}{2p_m}(1+\varepsilon_m), \qquad 0 \le \varepsilon_m \le (p_m-1)2^{-2^{2m-3}}. \tag{S3.1}$$

For a uniformly chosen orientation,

$$\Pr(\mathrm{Stab}_{\mathrm{GL}(2m,2)}(f) \neq C_3) \le (p_m-1)2^{-2^{2m-3}}. \tag{S3.2}$$

Consequently almost every orientation has stabilizer exactly C_3 as $m \to \infty$, and

$$a_m \sim \frac{3\,2^{(4^m-1)/3}}{2|\mathrm{GL}(m,4)|}.$$

Such orientations exist for every $m \ge 4$ and are nonlinear. Furthermore, exactly $a_m - 1$ of the a_m supported conjugacy classes are nonlinear for every $m \ge 2$.

Proof. A linear map giving cyclic orientations on the lines of $\mathcal{S}_m$ must be either $f_0(x) = \omega x$ or $f_0^2(x) = \omega^2 x$. On each line it has one of these two restrictions. Applying additivity to independent coordinate vectors and their sum forces their choices to agree, and the two $\mathbb{F}_2$-basis vectors on each coordinate line then determine the map on V. A spread stabilizer conjugates f_0 to one of these two maps. The centralizer of f_0 is $\mathrm{GL}(m,4)$; coordinatewise Frobenius $\sigma(x) = x^2$ conjugates f_0 to f_0^2. This proves the spread-stabilizer assertion.

Linear elements preserve the reference orientations; elements of $\mathrm{GL}(m,4)\sigma$ reverse all of them. Such an element sends an orientation of weight r to one of weight $N_m - r$. As N_m is odd, it fixes none. Scalars act trivially on the orientation set, giving the first equality in (S3.1) by Proposition S3.5. The orientation stabilizer is therefore the inverse image in $\mathrm{GL}(m,4)$ of its subset stabilizer in P_m.

For a nonscalar matrix A, its fixed projective points are the lines in its eigenspaces for the three nonzero elements of $\mathbb{F}_4$. If their dimensions are d_λ, the number is $\sum_\lambda (4^{d_\lambda}-1)/3$. Their dimensions sum to at most m and each is at most $m-1$. Convexity gives the upper bound $(4^{m-1}-1)/3+1$. Thus every nonidentity projective element moves at least $4^{m-1}-1$ points. Each nontrivial permutation cycle has at least two points, so

$$N_m - c(g) \ge \left\lceil \frac{4^{m-1}-1}{2} \right\rceil = 2^{2m-3}.$$

It fixes a proportion $2^{c(g)-N_m}$ of all orientations. Summing these proportions bounds both the nonidentity contribution in (S3.1) and, by the union bound, the probability in (S3.2). This is the classical motion argument [26] in the present projective action.

Finally $p_m < 2^{2m^2}/3$ and $2^{2m-3} \ge 2m^2$ for $m \ge 4$; the latter holds at $m = 4$ and increases thereafter. The exceptional bound is then less than one and tends to zero. The only two linear orientations have stabilizer $\mathrm{GL}(m,4)$, which is larger than C_3, so orientations with exactly C_3 are nonlinear. The two uniform orientations are conjugate by the semilinear Frobenius map, and no other orientation is linear; hence they constitute the single linear conjugacy class and the other $a_m - 1$ classes are nonlinear. □

For example $p_4 = 987033600$, and the exceptional proportion is at most $987033599/2^{32} < 0.230$. The estimate therefore already guarantees that more than 77% of the orientations on the standard eight-dimensional spread have exactly the scalar stabilizer. This conclusion concerns the standard spread family, without asserting a distribution over all spreads.

Corollary S3.9 (Six-dimensional standard-spread classes)**.** *The* 2^{21} *orientations of* $\mathcal{S}_3$ *give exactly* 80 *full* $\mathrm{GL}(6,2)$ *conjugacy classes supported on its underlying spread orbit. Classifying by the smaller of the two complementary orientation weights, the numbers are*

r	0	1	2	3	4	5	6	7	8	9	10
classes	1	1	1	2	3	5	8	10	13	17	19.

Only the weight-zero class is linear.

Proof. Here $|P_3| = 60480$. Its projective cycle types and multiplicities are listed below; a superscript is the number of cycles of that length.

matrix description	projective cycle type	multiplicity
scalar	1^{21}	1
$J_2(a) \oplus a$	$1^5\,2^8$	315
$J_3(a)$	$1\,2^2\,4^4$	3780
$aI_2 \oplus b$, $a \neq b$	$1^6\,3^5$	672
$J_2(a) \oplus b$, $a \neq b$	$1^2\,2^2\,3\,6^2$	10080
three distinct eigenvalues	$1^3\,3^6$	2240
quadratic block, normalized order 5	$1\,5^4$	8064
quadratic block, normalized order 15	$1\,5\,15$	16128
cubic block, projective order 3	3^7	1920
cubic block, projective order 7	7^3	5760
cubic block, projective order 21	21	11520

For completeness these multiplicities follow directly from rational canonical form. In $\mathrm{GL}(3,4)$, the first six types have respectively $3, 3, 3, 6, 6, 1$ conjugacy classes. Their corresponding centralizer orders are

$$181440, \quad 576, \quad 48, \quad 540, \quad 36, \quad 27.$$

Divide their total element counts by three to pass to projective matrices. For a quadratic block plus eigenvalue a, divide the quadratic roots by a. The 12 elements of $\mathbb{F}_{16} \setminus \mathbb{F}_4$ have orders 5 or 15, with 4 or 8 such roots. They give 6 or 12 matrix classes with centralizer order 45. For a cubic block, roots outside $\mathbb{F}_4$ have projective orders $3, 7, 21$, represented by $6, 18, 36$ field elements. They give $2, 6, 12$ matrix classes with centralizer order 63. For the Jordan types, $J_2(a) \oplus a$ has projective order two and five fixed points. The powers of $J_3(a)$ have respectively one, five, and twenty-one fixed points at exponents $1, 2, 4$, giving $1\,2^2\,4^4$. For $J_2(a) \oplus b$, the corresponding counts at exponents $1, 2, 3, 6$ are $2, 6, 5, 21$, giving $1^2\,2^2\,3\,6^2$. The remaining cycle types follow from the orders on the invariant line, quadratic component, or cubic field. All multiplicities sum to 60480, so the types are exhaustive.

Summing multiplicity times $2^{\text{number of cycles}}$ gives 9676800; (S3.1) gives $9676800/120960 = 80$. The number of r-subset classes under P_3 is the coefficient of z^r in

$$\frac{1}{60480} \sum_{g \in P_3} \prod_{C \text{ a point cycle of } g} (1 + z^{|C|}).$$

Using the table gives the displayed row. Semilinear conjugation pairs weights r and $21 - r$, so each entry also counts the oriented classes of the corresponding weight pair. Only the two uniform orientations are linear, and they belong to the same class. □

An independent complete traversal of the 2^{21} orientation vectors supplies all 80 representatives, their full linear stabilizer orders, and their orbit sizes. Among these, four classes have stabilizer C_3, accounting for 483840 orientations on the fixed standard spread. These representative and stabilizer data are finite computations, distinct from the general proof of Theorem S3.8. The supplied verifier

also independently enumerates the 60480 projective matrices and recovers the preceding cycle table and weight polynomial.

The general fixed-map description reduces counting to the line-spread enumeration problem; no general formula for s_n is supplied here. The affine orbit count is complete for every dimension, while the scalar Singer triples provide a distinguished family with controlled permutation cycle types and two-generator affine groups. Theorems 2.3, S3.7 and S3.8 give the general affine group structure, the full four-dimensional classification, and the standard-spread asymptotics. Corollary S3.9 counts all six-dimensional standard-spread oriented classes; it does not classify orientations on the other 131043 underlying spread types (out of 131044 in total).

S4 Nonlinear companion groups

An F-fixed map generates only a cyclic group of order three with its companion. Large companion groups therefore require a three-element F-orbit. If two companions both have cycle type $1+(2^n-1)$, their cycles have odd length, so their group is contained in A_{2^n}. The following example attains this upper bound in dimension four.

Proposition S4.1 (Two long cycles with distinct fixed points)**.** *Let Ω be a finite set of size $N \geq 4$, and suppose $f, g \in \mathrm{Sym}(\Omega)$ both have cycle type $1+(N-1)$. If their unique fixed points are distinct, then $H = \langle f, g \rangle$ acts doubly transitively on Ω. If in addition $N = 2^n$ with $n \geq 3$ and H contains a 3-cycle, then $H = A_N$.*

Proof. Let a be the unique fixed point of f. Since g fixes a different point, $g(a) \neq a$. The $(N-1)$-cycle f acts transitively on $\Omega \setminus \{a\}$, so H is transitive on Ω: its orbit containing a also contains $g(a)$ and every other point. Moreover $f \in H_a$, and f itself is transitive on $\Omega \setminus \{a\}$. Thus the point stabilizer is transitive on the remaining points and H is doubly transitive.

Now let $N = 2^n \geq 8$. An $(N-1)$-cycle is even, so $H \leq A_N$. The action is primitive since it is doubly transitive. The classical Jordan theorem for a primitive degree-N group containing a prime cycle of length $p \leq N-3$, applied with $p = 3$, implies $A_N \leq H$ (see, e.g., [17]). Therefore $H = A_N$. □

This sufficient condition separates the group-theoretic deduction from the difficult existence problem for complete mappings with prescribed companion cycle types: the distinct-fixed-point hypothesis gives double transitivity automatically, while the existence of a suitable short-cycle element still requires additional arguments.

Proposition S4.2 (A nonlinear alternating-group example)**.** *On $V = \mathbb{F}_2^4$, with integer labels and XOR addition, let f be the permutation in the first row below and let $g = F(f)$.*

x	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
$f(x)$	1	5	12	3	11	7	15	10	2	14	9	13	0	8	6	4
$g(x)$	3	0	5	10	1	13	11	9	14	6	8	15	12	7	2	4

Then f is complete and nonlinear. Each of $f, F(f), F^2(f)$ has cycle type $1+15$, with fixed points $3, 12, 1$ respectively, and

$$\langle f, F(f) \rangle = A_{16}.$$

Proof. The first row and $x \oplus f(x)$ each permute the labels $0, \ldots, 15$; inverting the latter gives the

second row. The cycles other than the indicated fixed points are

$$f : (0\ 1\ 5\ 7\ 10\ 9\ 14\ 6\ 15\ 4\ 11\ 13\ 8\ 2\ 12),$$
$$g : (0\ 3\ 10\ 8\ 14\ 2\ 5\ 13\ 7\ 9\ 6\ 11\ 15\ 4\ 1),$$
$$F^2(f) : (0\ 12\ 14\ 7\ 2\ 10\ 13\ 6\ 8\ 5\ 4\ 11\ 15\ 9\ 3).$$

The third cycle follows from $F^2(f) = \mathrm{id} \oplus f^{-1}$. An affine map would satisfy $f(1 \oplus 2) = f(1) \oplus f(2) \oplus f(0)$. Here these values are 3 and 8, so f is nonlinear. Since F preserves the affine class, all three companions are nonlinear.

Use right-to-left composition, write $[u, v] = uvu^{-1}v^{-1}$, and put $w = f^2 g$. Directly from the table,

$$w = (0\ 3\ 14)(1\ 5\ 2\ 10\ 12)(4\ 7\ 6\ 8\ 15\ 13\ 9), \qquad w(11) = 11.$$

Thus $s = w^{35} = (3\ 0\ 14)$. All generators are even, so $H = \langle f, g\rangle \le A_{16}$. The cycles $f^j s f^{-j} = (3\ f^j(0)\ f^j(14))$ belong to H, and their supports contain every point because f cycles all points other than 3.

To see that these cycles generate A_{16}, make a graph on the other 15 points, joining a, b whenever $(3\ a\ b)$ is available. There are no isolated vertices. If two graph components have edges ab and cd, the identity $[(3\ a\ b), (3\ c\ d)] = (3\ a\ c)$ joins them. After the graph is connected, conjugating $(3\ b\ c)$ by $(3\ a\ b)$ gives $(a\ 3\ c)$, so paths supply $(3\ a\ b)$ for every pair. Finally $(3\ a\ b)(3\ b\ c) = (a\ b\ c)$. These are all three-cycles, which generate A_{16}. Hence $H \ge A_{16}$, proving equality without a group-order search. □

Alternatively, the distinct fixed points 3 and 12 give double transitivity by Proposition S4.1; the explicit word w^{35} is a 3-cycle. The Jordan-theorem conclusion in that proposition therefore yields A_{16} directly. The elementary proof above is retained independently of Jordan's theorem.

Complete mapping polynomials over $\mathbb{F}_{16}$ and their reduced-degree distribution were computationally studied by Yuan–Tong–Zhang [28]; Bors–Wang studied cycle types and coset-wise affine constructions of complete mappings [22, 27]. These works provide important precedents for the finite-field maps and cycle-type methods. The result here concerns the additional group generated by one specified map and its prescribed companion, which is not the object counted by those cited results. A construction in infinitely many dimensions with simultaneous cycle-type and companion-group control is not supplied here. The groups generated by the full collections of complete mappings in [23] concern a different set of generators.

Proposition S4.3 (Translations of fixed maps)**.** *Let f_0 be an F-fixed complete mapping on $\mathbb{F}_2^n$ and let $b \neq 0$. Write $\tau_b(x) = x \oplus b$ and $f = \tau_b f_0$. Then f is complete and*

$$F(f) = f_0\tau_b, \qquad F^2(f) = \tau_b f_0 \tau_b, \qquad \langle f, F(f)\rangle = \langle f_0, \tau_b\rangle.$$

The orbit of 0 under this group is exactly the four-point line

$$U_b = \{0, b, f_0(b), f_0^2(b)\}.$$

The induced action on U_b is A_4. Thus this construction cannot give a transitive companion group when $n > 2$.

Proof. The fixed-map identities give $f_0^3 = \mathrm{id}$ and $\mathrm{id} \oplus f_0 = f_0^2$. Therefore $\mathrm{id} \oplus f = \tau_b f_0^2$ is a permutation, and its inverse is $g = f_0\tau_b$. Also

$$(\mathrm{id} \oplus g)(x) = x \oplus f_0(x \oplus b) = b \oplus f_0^2(x \oplus b),$$

so taking its inverse gives the formula for $F^2(f)$. Since $gf = f_0^2$, the group generated by f, g contains f_0 and then $\tau_b = f f_0^{-1}$; the reverse inclusion is immediate.

Proposition S3.3 makes U_b a two-dimensional subspace, invariant under both f_0 and τ_b. Starting at 0, translation reaches b and f_0 reaches the other two nonzero points, so the orbit is all of U_b. On these points $f_0 = (b\ f_0(b)\ f_0^2(b))$ and $\tau_b = (0\ b)(f_0(b)\ f_0^2(b))$. These generate the three double transpositions and a three-cycle, hence A_4. □

S5 Additional scalar consequences

S5.1 The exact index-three threshold

Corollary S5.1 (The exact threshold at index three)**.** *For even $n \geq 4$, the specified subgroup*

$$\mathbb{F}_{2^n}^{+} \rtimes C_{(2^n-1)/3}$$

is primitive and is generated by a scalar affine complete mapping and its prescribed companion if and only if $n \geq 8$. Thus index-three proper primitive companion groups exist in every even dimension from eight onwards.

Proof. For every even $n \geq 4$, writing $t = 2^{n/2} > 2$ gives $(t^2-1)/3 > t-1$. The field-degree argument in Theorem 6.2 proves primitivity. At $n = 4$ the complement has order $5 = 2^2 + 1$, excluded by Theorem 6.5. At $n = 6$ it has order 21, and the polynomial identities following Corollary 2.6 give $\mathcal{R}_{21} = \varnothing$.

For $k = 3$, $b_3 = 2/9$. Using $\omega(d) < 2n/3$, the uniform lower bound in (6.3) is greater than

$$\frac{2^n}{12} - \frac{8}{9} - \left(\frac{2n}{3} + \frac{2}{9}\right) 2^{n/2}.$$

This expression equals $424/3 > 0$ at $n = 14$. After division by $2^{n/2}$, increasing n by two doubles its positive exponential term, whereas its linear term increases by $4/3$ and its negative exponential term decreases in magnitude. Its increase is greater than $2^{n/2}/12 - 4/3 > 0$ for even $n \geq 14$, proving positivity there. For $n = 10$, $d = 341 = 11 \cdot 31$ and $\omega(d) = 2$; for $n = 12$, $d = 1365 = 3 \cdot 5 \cdot 7 \cdot 13$ and $\omega(d) = 4$. The bound $L_{q,3,d}$ equals $40/3$ and $632/9$, respectively, so these dimensions are covered as well.

Finally take $n = 8$ and

$$\mathbb{F}_{256} = \mathbb{F}_2[t]/(m(t)), \qquad m(t) = t^8 + t^4 + t^3 + t + 1, \qquad a = t^5 + t^3 + t^2 + t.$$

The irreducibility criterion for m is certified by $t^{256} \equiv t \pmod{m}$ and $\gcd(m, t^{16} + t) = 1$; these are polynomial identities over $\mathbb{F}_2$. Direct reduction modulo m gives

$$a^{85} = (1+a)^{85} = 1, \qquad a^5 = t^7 + t^5 + t^3 + t + 1 \neq 1, \qquad a^{17} = t^7 + t^6 + t^5 + t^3 + t^2 + 1 \neq 1.$$

Since $85 = 5 \cdot 17$, these identities show that a has exact order 85 and $1 + a \in C_{85}$. Thus $a \in \mathcal{R}_{85}$; $f(x) = ax + c$ realizes the group for every $c \neq 0$ by Corollary 2.5. This closes all the even dimensions claimed. □

The square-root estimate in main Proposition 6.1 is classical; the results specialize it to the subgroup generated by the linked slopes a and $1 + a$. The sufficient bound is not a general necessary condition for realizability, and no priority claim is made for the character-sum method or these particular applications.

S5.2 Primitive classification and generating triangles

Proposition S5.2 (Primitive inverse classification and linear conjugacy)**.** *Up to* $\mathrm{GL}(n,2)$ *conjugacy, the primitive groups generated by an affine complete mapping and its prescribed companion are exactly*

$$H_d = \mathbb{F}_{2^n}^+ \rtimes C_d, \qquad d \mid 2^n - 1, \quad \mathrm{ord}_d(2) = n, \quad \mathcal{R}_d \neq \varnothing.$$

Distinct admissible d *give distinct group-conjugacy classes. For each such* d*, the normalizer in* $\mathrm{GL}(n,2)$ *has order* $n(2^n-1)$ *and is the semilinear scalar group* $\Gamma\mathrm{L}(1,2^n)$*. The affine complete mappings realizing a linear conjugate of* H_d *form exactly* $|\mathcal{R}_d|/n$ *linear-conjugacy classes, each of size* $|\mathrm{GL}(n,2)|$*.*

Proof. Corollary 2.4 makes the linear part A of every primitive realization irreducible and its translation nonzero. The field $\mathbb{F}_2[A]$ then identifies V with $\mathbb{F}_{2^n}$; the complement is generated by a and $1+a$ and has some order d. The characterization follows from Corollary 2.5. Conversely that corollary realizes every listed H_d. Different orders $2^n d$ distinguish the group classes.

A linear transformation normalizes H_d exactly when it normalizes its stabilizer C_d of zero. Since $\mathbb{F}_2[C_d] = \mathbb{F}_{2^n}$, conjugation by such a transformation induces an automorphism of this field, hence a power of Frobenius. After removing that power, the transformation centralizes all scalar multiplications and is itself scalar. Conversely every map $x \mapsto bx^{2^i}$ normalizes C_d. The normalizer therefore has the asserted order.

For maps realizing the specified H_d, normalizer conjugation sends (a,c) to (a^{2^i}, bc^{2^i}). Scalar transformations act transitively on $c \neq 0$. Every $a \in \mathcal{R}_d$ has degree n over $\mathbb{F}_2$: a proper subfield containing a would contain $1+a$ and all of C_d. Thus its Frobenius orbit has length n, giving $|\mathcal{R}_d|/n$ normalizer orbits. Conjugacy between two maps with group H_d must normalize H_d, so these are also the full linear-conjugacy classes when group conjugates are included. Finally, a linear transformation centralizing $f(x) = ax + c$ centralizes the irreducible multiplier a, so is scalar; fixing $c \neq 0$ forces it to be the identity. All full linear-conjugacy classes consequently have size $|\mathrm{GL}(n,2)|$. □

For an explicit polynomial encoding, set

$$G_e(t) = \gcd(t^e + 1, (t+1)^e + 1), \qquad P_d(t) = \prod_{e \mid d} G_e(t)^{\mu(d/e)}.$$

Although written as a quotient, P_d is a monic squarefree polynomial over $\mathbb{F}_2$ with root set $\mathcal{R}_d$. Indeed the exponent of a root whose pair generates C_h is $\sum_{h \mid e \mid d} \mu(d/e)$, which is one if $h = d$ and zero otherwise. When $\mathrm{ord}_d(2) = n$, all irreducible factors of P_d have degree n and encode the classes in Proposition S5.2. For example the index-three counts in dimensions $8, 10, 12, 14$ give $3, 9, 33, 123$ map classes, respectively.

The graph interpretation clarifies the distinction between subgroup membership and joint generation. Write

$$\Gamma(q,d) = \mathrm{Cay}(\mathbb{F}_q^+, C_d), \qquad x \sim y \iff y - x \in C_d.$$

This is the usual generalized Paley graph (including the complete graph when $d = q-1$). Its connectivity and ordinary triangle counts are classical [4, 5]. For a triangle with ordered vertices (x,y,z), define its ratio subgroup to be

$$K(\{x,y,z\}) = \langle a, 1+a \rangle, \qquad a = \frac{z-x}{y-x}.$$

This subgroup is independent of the vertex ordering. We call the triangle *generating* when its ratio subgroup is C_d. The definition uses normalized ratios, not the subgroup generated by the unnormalized edge differences.

Proposition S5.3 (Generating triangles and companion realization)**.** *For $d > 1$ dividing $q - 1$, $\Gamma(q, d)$ is connected exactly when $\mathrm{ord}_d(2) = n$. Each edge belongs to T_d ordinary triangles and $|\mathcal{R}_d|$ generating triangles. The total number of generating triangles is $qd|\mathcal{R}_d|/6$. Thus H_d is realizable exactly when the graph is connected and has a generating triangle. If $d > 3$, the action of H_d on generating triangles is free, with $|\mathcal{R}_d|/6$ orbits; in particular $6 \mid |\mathcal{R}_d|$.*

Proof. The additive span of C_d is closed under products, contains one, and is a finite subfield. It is therefore $\mathbb{F}_2[C_d] = \mathbb{F}_{2^{\mathrm{ord}_d(2)}}$. The components are its additive cosets, proving the connectivity assertion (compare [4, Theorem 2.2]).

For the edge $\{0, 1\}$, its third vertices are precisely the a with $a, 1 + a \in C_d$, and the generating ones are precisely $a \in \mathcal{R}_d$. Reordering vertices replaces a by one of

$$a, \quad 1 + a, \quad a^{-1}, \quad (1 + a)^{-1}, \quad \frac{a}{1 + a}, \quad \frac{1 + a}{a};$$

these all generate the same subgroup together with one plus themselves. Translations and multiplication by C_d preserve the ratios and act transitively on edges. Counting the $qd/2$ edges and three edges per triangle gives the total. The realization equivalence now follows from Corollary 2.5.

For $d > 3$ the six transforms of $a \in \mathcal{R}_d$ are distinct. A transposition can fix only an excluded value 0 or 1, whereas a three-cycle can fix only $a^2 + a + 1 = 0$, whose pair generates C_3. Any affine stabilizer of a generating triangle induces a vertex permutation and would make the corresponding transform fix a. It must therefore fix the vertices individually, forcing its multiplier to be one and its translation to be zero. The action is free. Each orbit has size qd, giving the orbit count and divisibility. No claim about the full graph automorphism group is required here. □

In the family of Corollary 6.7, the graph is connected and every edge has exactly $t - 2$ third vertices. For an edge $\{x, y\}$ they are $x + (y - x)a$ with $a \in \mathbb{F}_t \setminus \{0, 1\}$. Thus all its triangles lie in the clique $x + (y - x)\mathbb{F}_t$, and none is generating. These graphs are the already studied $(t + 1)$-power graphs $\Gamma_{t,4}(1)$ of Podestá and Videla [6]; their ordinary intersection parameter $t - 2$ follows also from [6, Theorem 5.2]. The descent proof supplies the exact subfield location and its companion-group obstruction. Connectedness and ordinary triangles alone do not guarantee realization. The graph family, character-sum method, and ordinary triangle counts are not claimed as new, and priority of the companion-specific applications remains to be established.

S6 Single full-order slopes and fiber symmetry

In this section use the quartic notation of main Section 3: $t = 2^s$, $A = t + 1$, $B = t^2 + 1$, $d = AB$, $K = \mathbb{F}_t \subset E = \mathbb{F}_{t^2} \subset L = \mathbb{F}_{t^4}$, $U = \ker N_{E/K}$ and $V = \ker N_{L/E}$. The main paper develops the mixed-character estimate directly. Here we retain the separate single-character argument, the four-torsion fiber symmetry and the single full-order existence proof. These give a second route to the weaker one-slope conclusion.

S6.1 A single-character estimate

Lemma S6.1 (Uniform character bound on a norm fiber)**.** *Let b be as in Lemma 3.2, and let $u \in U$ be the unique square root of b. For every nontrivial multiplicative character χ of V, one has*

$$\left| \sum_{a \in X_b(K)} \chi(a/u) \right| \le 8\sqrt{t}. \tag{S6.1}$$

Proof. We use the classical Weil bound for a geometrically nontrivial finite-order rank-one character on a curve: on a genus-g curve, tame ramification at r geometric points gives the bound $(2g-2+r)\sqrt{t}$, when the character is nontrivial at every one of those points. This is the curve Artin-character estimate discussed in [8, §4 of the author's version]; the Lang-torsor interpretation of finite-field multiplicative characters is also used in [7]. For completeness, the dimension underlying this estimate is $2g-2+r$: compactly supported cohomology in degrees zero and two vanishes, and the tame Euler-characteristic formula gives this dimension in degree one. The trace formula and the Weil theorem for curves bound each eigenvalue by $\sqrt{t}$. For finite-order characters this is a character component of a finite cover of the curve. We now check all these hypotheses here.

Let $\mathcal{T} = \mathrm{Res}_{E/K}(\ker N_{L/E})$ denote the two-dimensional norm torus. It has $\mathcal{T}(K) = V$, and $a \mapsto a/u$ maps X_b to $\mathcal{T}$, since $N_{L/E}(u) = u^2 = b$. Over $\overline{K}$ use coordinates (z_0, z_1) on the torus; its Frobenius is

$$F_{\mathcal{T}}(z_0, z_1) = (z_1^{-t}, z_0^t).$$

The Lang isogeny $z \mapsto F_{\mathcal{T}}(z)/z$ has constant kernel V. Pulling it back along $a \mapsto a/u$, and writing $r_0 = x/u$, $r_1 = y/u^t$, gives

$$\frac{z_1^{-t}}{z_0} = r_0, \qquad \frac{z_0^t}{z_1} = r_1, \qquad \text{hence} \qquad z_0^B = \frac{r_1^t}{r_0}.$$

Thus this pullback is geometrically a Kummer cover with defining function $h = (y/u^t)^t/(x/u) = y^t/x$, since $u^{t^2} = u$. For a character of exact order $m \mid B$, its quotient cover has equation $z^m = h$ over $\overline{K}$. This geometric Kummer presentation must be distinguished from the K-structure of the quotient cover: the latter is inherited from the pullback of the K-defined Lang isogeny, with its corresponding Frobenius descent datum. In particular, h need not belong to the function field $K(C_b)$ in these split coordinates. All statements about geometric ramification and geometric connectedness are checked after base change to $\overline{K}$, whereas the character sum is taken using the descended K-cover.

The Frobenius descent can be made explicit, so that the character trace is not merely inferred from a convention. On geometric points, the K-Frobenius on the base and on the pullback Lang cover is

$$\begin{aligned} F_{C_b}(x, y) &= \left(\frac{b}{y^t}, x^t\right), \\ F_Y(x, y, z_0) &= \left(\frac{b}{y^t}, x^t, \frac{x}{u} z_0\right). \end{aligned} \tag{S6.2}$$

Indeed, the two Lang equations give $z_1^{-t} = r_0 z_0$, while $F_{\mathcal{T}}$ acts on the first split coordinate as $z_0 \mapsto z_1^{-t}$. The displayed formula is also compatible with the Kummer equation, since $u^B = b$ and

$$h\left(\frac{b}{y^t}, x^t\right) = \frac{x^{t^2} y^t}{b} = \left(\frac{x}{u}\right)^B h(x, y).$$

For a K-rational point, $(x, y) = (a, a^t)$ with $a^B = b$; thus Frobenius fixes the base point and acts on its geometric Lang fiber by $z_0 \mapsto (a/u) z_0$. This is precisely the deck transformation indexed by $a/u \in V$. In the character eigensheaf associated with χ, its Frobenius trace is therefore $\chi(a/u)$, or $\chi^{-1}(a/u)$ if the inverse Frobenius convention is used. The latter choice is immaterial for the asserted bound. In particular, this computation verifies the Frobenius trace identification for the actual K-descent of the cover, rather than for the bare geometric Kummer equation.

The zeros and poles of x are simple and supported on D_x; those of y are simple and supported

on the disjoint set D_y. The valuations can be read point by point:

points	$\mathrm{ord}_P(h)$
$(0,1),\ (0,b^{-1})$	-1
$(\infty,1),\ (\infty,b^{-1})$	1
$(1,0),\ (b,0)$	t
$(1,\infty),\ (b,\infty)$	$-t$

Thus the eight nonzero valuations are $1,-1,t,-t$, each twice. Since m is odd and $\gcd(m,t)=1$, the character is tamely and nontrivially ramified at all eight points. A valuation of absolute value one also makes this cover geometrically connected, excluding a constant-field character or a geometrically trivial pullback. The conductor degree is consequently exactly eight, independently of m,t,b. Lemma 3.2 and the curve bound now give $(2\cdot 1-2+8)\sqrt{t}=8\sqrt{t}$. □

S6.2 The four-torsion symmetry

Proposition S6.2 (Four-torsion symmetry of a norm fiber)**.** *In the setting of Lemma 3.2, let $u\in U$ satisfy $u^2=b$. On the geometric (x,y)-model of C_b set*

$$\rho(x,y)=\left(\frac{1}{y},\frac{x}{b}\right),\qquad h=\frac{(y/u^t)^t}{x/u}=\frac{y^t}{x}.$$

Then ρ descends to a K-automorphism of order four. After choosing a K-rational origin, it is translation by a K-rational point of exact order four. The eight boundary points form exactly two ρ-orbits of length four:

$$(0,1)\longmapsto(1,0)\longmapsto(\infty,b^{-1})\longmapsto(b,\infty)\longmapsto(0,1),$$
$$(0,b^{-1})\longmapsto(b,0)\longmapsto(\infty,1)\longmapsto(1,\infty)\longmapsto(0,b^{-1}).$$

Moreover, as identities in the geometric function field,

$$\rho^*h=(uy)^Bh^{-t},\qquad(\rho^2)^*h=h^{-1}. \tag{S6.3}$$

For every character χ of V, the sums in Lemma S6.1 satisfy

$$S_b(\chi):=\sum_{a\in X_b(K)}\chi(a/u)=S_b(\chi^{-t})=S_b(\chi^{-1})\in\mathbb{R}.$$

Every nontrivial character belongs to an orbit of length four under $\chi\mapsto\chi^{-t}$. If N_b^{prim} denotes the number of $a\in X_b(K)$ with $\mathrm{ord}(a)=d$, then $4\mid\#X_b(K)$ and $4\mid N_b^{\mathrm{prim}}$.

Proof. The field automorphism $\sigma:a\mapsto a^t$ of L/K extends as $\sigma\otimes 1$ to an R-linear automorphism of $L\otimes_K R$ for every K-algebra R; composing this extension with inversion gives a K-automorphism of the ambient restriction-of-scalars torus. On K-points its action is $a\mapsto(a^t)^{-1}$. Here this notation refers to the Galois automorphism σ, not the relative Frobenius morphism on the torus. The functorial norm identities

$$N_{L/E}((\sigma a)^{-1})=b^{-t}=b,\qquad N_{L/K}(1+(\sigma a)^{-1})=\frac{N_{L/K}(1+a)}{N_{L/K}(a)}=1$$

show that it preserves both defining conditions, since $N_{L/K}(a)=N_{E/K}(b)=1$. It gives the displayed ρ in split coordinates. Equivalently, the split-coordinate K-Frobenius descent is $(x,y)\mapsto(b/y^t,x^t)$, with which ρ commutes. Direct iteration gives

$$\rho^2(x,y)=(b/x,b^{-1}/y),\qquad\rho^4=1.$$

The eight boundary points are paired without fixed points by ρ^2. Any remaining geometric fixed point would have $x^2 = b$, $y^2 = b^{-1}$, hence $(x, y) = (u, u^{-1})$. Substitution in the equation of C_b forces $b + b^{-1} = 1$, equivalently $u^2 + u + 1 = 0$, whence $b^3 = 1$, contrary to $b^3 \neq 1$. Thus ρ^2, and hence ρ, has no geometric fixed point. The bound $\#C_b(K) \geq t + 1 - 2\sqrt{t} > 0$ supplies a rational origin. After choosing a rational origin, any automorphism of a genus-one curve is a translation followed by an origin-preserving automorphism. If the latter is nontrivial, its difference from the identity is a nonzero, hence surjective, elliptic-curve endomorphism, so the composite has a geometric fixed point. Consequently ρ is a translation, necessarily by a K-rational point of exact order four.

The two boundary cycles follow by substitution in the rational formula, interpreted on $\mathbb{P}^1 \times \mathbb{P}^1$. Since $u^{t+1} = 1$, $u^B = u^2 = b$ and $b^t = b^{-1}$,

$$\rho^* h = \frac{(x/b)^t}{1/y} = bx^t y = (uy)^B \left(\frac{y^t}{x}\right)^{-t},$$

while the formula for ρ^2 yields $(\rho^2)^* h = h^{-1}$. For $a = uv$ with $v \in V$, ρ sends a to uv^{-t}. It therefore permutes $X_b(K)$ and preserves $\mathrm{ord}(a)$; changing variables in the character sum proves the stated equalities. Finally $B = t^2 + 1$ is odd, $(-t)^2 \equiv -1 \pmod{B}$. For a character of order $m > 1$, an orbit of length one or two would imply $m \mid 2$, which is impossible since $m \mid B$ is odd. Thus every nontrivial character has an orbit of length four. Translation by a point of exact order four acts freely, proving both divisibilities. □

These symmetries do not by themselves reduce the bound in Lemma S6.1: grouping the characters into four-element orbits leaves the total absolute sieve coefficients unchanged. The estimate used below remains $8\sqrt{t}$.

S6.3 The single full-order statistic

Theorem S6.3 (Primitive slopes in the inner critical family)**.** *Put $\theta(B) = \varphi(B)/B$ and $W(B) = 2^{\omega(B)}$. For each $b \in U$ of order A, let*

$$N_b^{\mathrm{prim}} = \#\{a \in X_b(K) : \mathrm{ord}(a) = d\}.$$

Then

$$\left|N_b^{\mathrm{prim}} - \theta(B)(t+1)\right| \leq \theta(B)(8W(B) - 6)\sqrt{t}. \tag{S6.4}$$

In particular, with $N_d^{\mathrm{prim}} = \#\{a \in C_d : \mathrm{ord}(a) = d,\ 1 + a \in C_d\}$,

$$\left|N_d^{\mathrm{prim}} - \varphi(A)\theta(B)(t+1)\right| \leq \varphi(A)\theta(B)(8W(B) - 6)\sqrt{t}. \tag{S6.5}$$

Thus $N_d^{\mathrm{prim}} \sim \varphi(A)\theta(B)t$ as $s \to \infty$.

Proof. Every $a \in X_b(K)$ has a unique expression $a = uv$ with $v \in V$, where u is the square root of b in U. Since $\mathrm{ord}(u) = A$ and $\gcd(A, B) = 1$, it has order d exactly when v has order B. The usual character indicator in a cyclic group of order B is

$$\mathbf{1}_{\mathrm{ord}(v)=B} = \theta(B) \sum_{m \mid B} \frac{\mu(m)}{\varphi(m)} \sum_{\mathrm{ord}(\chi)=m} \chi(v).$$

Indeed the sum factors over primes $p \mid B$: the corresponding factor is zero on V^p and $p/(p-1)$ off V^p, which cancels the factor $(p-1)/p$ in $\theta(B)$. The trivial-character term is $\theta(B)\#X_b(K)$.

For each nontrivial squarefree divisor m the sum of the absolute values of its coefficients is one. There are $W(B)-1$ such divisors. Lemmas 3.2 and S6.1 therefore give an error of at most $\theta(B)(2+8(W(B)-1))\sqrt{t}$, proving (S6.4).

An element of order d has relative norm of order A, because $N_{L/E}(a)=a^B$. Hence these $\varphi(A)$ norm fibers partition the elements counted by N_d^{prim}. Summing proves (S6.5). Finally $2^{\omega(N)}=N^{o(1)}$: for any $\epsilon>0$, all but finitely many primes satisfy $2\le p^{\epsilon}$, and the remaining primes contribute only a constant. Since $B\asymp t^2$, the relative error tends to zero. □

Theorem S6.4 (All-dimensional realization at index $t-1$)**.** *For every $s\ge 2$, there exists $a\in\mathbb{F}_{2^{4s}}$ with*

$$\operatorname{ord}(a)=(2^s+1)(2^{2s}+1),\qquad (1+a)^{(2^s+1)(2^{2s}+1)}=1.$$

Consequently $\mathcal{R}_d\neq\varnothing$ at $k=t-1$ for every such s. The proper primitive group $L^+\rtimes C_d$ is generated by $f(x)=ax+c$ and $F(f)$ for every $c\in L^\times$.

Proof. Every prime divisor p of $B=t^2+1$ is 1 modulo 4, since $t^2\equiv -1 \pmod p$. If $p\ge 256$, then $2\le p^{1/8}$. For the smaller primes a direct integer product gives

$$\prod_{\substack{p<256 \text{ prime}\\ p\equiv 1 \pmod 4}} \frac{256}{p} < 32^8.$$

The 24 primes in this product and the exact integer comparison are supplied with the finite verifier in Appendix B. It follows for every B under consideration that

$$W(B)<32B^{1/8}.$$

For $s\ge 33$, $t^2\ge 2^{66}$ and $t^2+1<2t^2$ imply $t^4>2^{64}(t^2+1)$. Equivalently, $\sqrt{t}>256B^{1/8}>8W(B)$. The lower bound in (S6.4) is then positive for every primitive $b\in U$.

For $2\le s\le 32$, Appendix B gives explicit field towers $E=\mathbb{F}_2[X]/(M_s)$ and $L=E[Z]/(Z^2+cZ+b)$, in which $a=Z$ is a certificate. Irreducibility, both norm conditions and the exact order d are verified by deterministic finite arithmetic, with complete factorizations and recursively certified primes. These 31 cases exhaust the remaining range; they are not an extrapolation.

Finally $d>t^2-1$ implies $\mathbb{F}_2[a]=L$: every proper subfield of L has at most t^2 elements. Thus C_d acts irreducibly, while $k=t-1>1$ makes it a proper multiplier group. The affine conclusion follows from Corollary 2.5. □

The two adjacent indices $t-1$ and $t+1$ therefore have opposite realization behavior for every $s\ge 2$. This does not settle the universal question for all admissible $k\le q^{1/4}$. Also, N_d^{prim} counts primitive individual slopes; it is a subset count within $|\mathcal{R}_d|$. The joint and double full-order counts in the main paper use the mixed-character estimate. Ordinary intersections for index $t-1$ belong to cyclotomic number theory, including the general norm-based treatment of Kudo and Nogata [16]. The additional exact-order condition is the focus here. The classical Weil bound, Lang construction, and primitive-character sieve are attributed above; priority for this particular realization theorem is not asserted.

S7 The trace-one curve and its Lang covers

Let $t=2^s$, $s\ge 2$, $K=\mathbb{F}_t$, $E=\mathbb{F}_{t^2}$, $L=\mathbb{F}_{t^4}$, $A=t+1$, $B=t^2+1$ and $d=AB$. Write $U=\ker N_{E/K}$ and $V=\ker N_{L/E}$. The main paper proves double-order existence in every primitive inner-norm

fiber in Theorem 4.5. The complete relative-trace classification and two off-slice certificates provide another proof of full-family existence. We retain here a separate geometric proof for $c = 1$, also analyzing the parameter excluded by the good-fiber hypothesis $b^3 \neq 1$ in Proposition 3.4. It specifies the Lang multipliers and all four inertia characters at this parameter. Combined with the wider field-certificate range, it provides an alternative existence route; the sharper trace-one positivity classification remains Theorem 5.4.

Proposition S7.1 (A trace-one double full-order subfamily)**.** *For $t = 2^s$, $s \geq 2$, use $A = t + 1$, $B = t^2 + 1$ and $d = AB$ as above. Put*

$$\Delta_t = \#\{\gamma \in V : \tau = \gamma + \gamma^{-1} \in U,\ \mathrm{ord}(\gamma) = B,\ \mathrm{ord}(\tau) = A\}.$$

Every element counted by Δ_t gives, injectively,

$$a = (1 + \gamma)^{-1} \in C_d, \qquad 1 + a = a^{t^2}, \qquad \mathrm{ord}(a) = \mathrm{ord}(1 + a) = d. \tag{S7.1}$$

In particular $D_d \geq \Delta_t$. For any characters η of U and χ of V with $(\eta, \chi) \neq (1, 1)$,

$$\left| \sum_{\substack{\gamma \in V \\ \gamma + \gamma^{-1} \in U}} \eta(\gamma + \gamma^{-1}) \chi(\gamma) \right| \leq 4\sqrt{t}. \tag{S7.2}$$

Consequently

$$|\Delta_t - \theta(d)(t + 1)| \leq \theta(d)(4W(d) - 2)\sqrt{t}. \tag{S7.3}$$

The sufficient condition $\sqrt{t} \geq 4W(d)$ implies $\Delta_t > 0$; it holds for every $s \geq 33$ by the integer bounds in Appendix B.

Proof. The full-order construction.

Here $\gamma^{t^2} = \gamma^{-1}$ and $\gamma \neq 1$. Thus $a^{t^2} = \gamma/(1 + \gamma) = 1 + a$. Also

$$N_{L/E}(1 + \gamma) = \gamma + \gamma^{-1} = \tau, \qquad a^B = \tau^{-1}, \qquad a^{t^2 - 1} = \gamma.$$

Since $N_{E/K}(\tau) = 1$, the norm of a to K is one, so $\mathrm{ord}(a)$ divides d. The two displayed powers force both A and B to divide this order. Their gcd is one, and hence the order is d. The conjugate $1 + a$ has the same order. The formula for a is injective in γ.

The curve and its four boundary points.

We analyze the norm fiber X_1 directly, rather than applying Proposition 3.4, whose good-fiber hypothesis excludes this parameter. Write its variable as γ. Over $\overline{K}$, the relative norm-one torus has coordinates (x, y, x^{-1}, y^{-1}). Its K-points have $x = \gamma$, $y = \gamma^t$, $\gamma \in V$, and its Frobenius is

$$F_Y(x, y) = (y^{-t}, x^t).$$

The condition $N_{E/K}(\gamma + \gamma^{-1}) = 1$ gives the projective closure $Y_{\overline{K}} \subset \mathbb{P}^1 \times \mathbb{P}^1$ of

$$(x + 1)^2 (y + 1)^2 = xy. \tag{S7.4}$$

Here Y denotes the K-curve with the inherited torus descent F_Y, not the split curve equipped with coordinatewise t-power Frobenius. The descent extends to the projective closure and preserves its boundary. In particular the relation $y = x^t$ is imposed on K-points, not in its geometric function

field. On the affine chart the two partial derivatives are y and x; their only common zero $(0,0)$ is not on the curve. The boundary is

$$P_0 = (0,1), \quad P_\infty = (\infty,1), \quad Q_0 = (1,0), \quad Q_\infty = (1,\infty).$$

At P_0, with $u = y+1$, the equation is $(1+x)^2u^2 + x(1+u) = 0$, whose derivative with respect to x is one at the point. Thus it is smooth and $v_{P_0}(x) = 2$. Reciprocating x or y, and interchanging them, gives the same smoothness test at the other boundary points. This proves that the bidegree-$(2,2)$ curve is smooth. It is geometrically irreducible. Smoothness implies that the curve is reduced and that distinct irreducible components would be disjoint. Every component has a nonzero effective bidegree (a_i, b_i), with $a_i, b_i \geq 0$ and $\sum_i (a_i, b_i) = (2,2)$. If one component has both entries positive, its intersection $a_i b_j + a_j b_i$ with every other component is positive. If no component has both entries positive, the sum $(2,2)$ forces a pure vertical component and a pure horizontal component, whose intersection is again positive. Both possibilities contradict disjointness. Adjunction on $\mathbb{P}^1 \times \mathbb{P}^1$ now gives genus $(2-1)(2-1) = 1$. Frobenius permutes the four boundary points in the cycle

$$P_0 \longmapsto Q_0 \longmapsto P_\infty \longmapsto Q_\infty \longmapsto P_0.$$

There is no K-rational boundary point. Therefore the open curve Y° has exactly the points in the sum (S7.2), and $|\#Y^\circ(K) - (t+1)| \leq 2\sqrt{t}$ by Hasse–Weil.

Actual Lang covers and their character traces.

Let $\mathcal{U} = \ker N_{E/K}$ and $\mathcal{V} = \mathrm{Res}_{E/K}(\ker N_{L/E})$; their K-point groups are U and V. On Y° use the maps $\tau = \gamma + \gamma^{-1}$ to $\mathcal{U}$ and γ to $\mathcal{V}$ itself. Write

$$w_0 = (x+1)^2/x, \qquad h = y^t/x.$$

The split coordinates of τ are (w_0, w_0^{-1}). The two Lang pullbacks, with their inherited K-structures, are

$$z_U^A = w_0^{-1}, \qquad z_V^B = h. \tag{S7.5}$$

For the first, the Lang equation is $z_U^{-t}/z_U = w_0$. For the second, the split equations are $z_1^{-t}/z_0 = x$, $z_0^t/z_1 = y$; elimination gives $z_0^B = y^t/x$, with $z_V = z_0$. Thus the actual functions are w_0^{-1} and h, not merely a geometrically convenient Kummer model. Above F_Y, Frobenius on the two covering coordinates is

$$(z_U, z_V) \longmapsto (w_0 z_U, x z_V).$$

This follows from the split Lang equations. It also respects (S7.5), since $w_0(F_Y) = w_0^{-t}$ and $h(F_Y) = x^B h$. At a K-point the deck multipliers are exactly τ and γ. Choosing the character convention for these multipliers gives trace $\eta(\tau)\chi(\gamma)$; the inverse convention just inverts both characters.

Geometric connectedness and conductor.

The complete boundary valuation table is

	w_0	h
P_0	-2	-2
P_∞	-2	2
Q_0	2	$2t$
Q_∞	2	$-2t$

(S7.6)

For example $x+1$ has simple zeros at Q_0, Q_∞ and a double pole at P_∞; y has a double zero at Q_0 and a double pole at Q_∞. These observations give every entry of the table. Suppose for $0 \leq e_0 < A$, $0 \leq e_1 < B$ that

$$(w_0^{-1})^{Be_0} h^{Ae_1} \in \overline{K}(Y)^{\times AB}.$$

Its valuation at P_0 is $2Be_0 - 2Ae_1$. Reduction modulo A and modulo B, using that A, B are odd and coprime, gives $A \mid e_0$ and $B \mid e_1$. This single valuation separates the two cyclic components because reduction modulo each order kills the other component. Thus $e_0 = e_1 = 0$. The Kummer criterion proves that the product cover in (S7.5) is geometrically connected with full group $U \times V$; in particular every nontrivial character quotient is geometrically nontrivial.

All cover degrees are odd, so the ramification is tame. At every boundary point the valuation of w_0^{-1} is one of ± 2, and that of h is one of ± 2 or $\pm 2t$. Since A and B are odd, $\gcd(2, A) = \gcd(2t, B) = 1$; each coordinate therefore generates its cyclic deck factor. A nontrivial character on either factor is nontrivial on that coordinate, and the two character values cannot cancel because their orders divide the coprime integers A and B. Thus every nontrivial pair is ramified at each of the four boundary points, with Swan conductor zero and local Artin conductor one. Its total conductor is exactly four, and it is unramified on Y°. The associated rank-one finite-monodromy sheaf is pure of weight zero and geometrically nontrivial. Thus $H_c^0 = H_c^2 = 0$, and Grothendieck–Ogg–Shafarevich gives $\dim H_c^1 = 2g(Y) - 2 + 4 = 4$. The classical curve character bound [8, §4, Lemma 1 in the author's version], with the Lang construction as in [7], bounds each Frobenius eigenvalue by $\sqrt{t}$. This proves (S7.2).

The two primitive indicators and positivity.

Apply the primitive-character indicator to $\tau \in U$ and $\gamma \in V$. Their product has constant coefficient $\theta(A)\theta(B) = \theta(d)$ and total absolute coefficient sum $\theta(d)W(A)W(B) = \theta(d)W(d)$. The trivial pair has Hasse error $2\theta(d)\sqrt{t}$. The other pairs have total coefficient mass $\theta(d)(W(d) - 1)$ and character bound $4\sqrt{t}$. Their sum is exactly the error in (S7.3). If $\sqrt{t} \geq 4W(d)$, then $(t + 1) - (4W(d) - 2)\sqrt{t} > 0$, proving positivity. The independent integer bound in main Lemma B.1 supplies this condition for every $s \geq 33$, proving $\Delta_t > 0$ throughout that range. □

Using the classical reciprocal-trace substitution [18, §4], the argument above verifies the descended norm geometry and four-point conductor for this specialization.

S.A Self-contained proof for the even-characteristic case

Proof of Theorem S2.2. The statement is classical (see [19] and [20, Theorem 1, p. 1]; it also follows from the equivalent square-order formulation in [18, Theorem 1.5, p. 170]); we give a self-contained proof in the even-characteristic case used above. Cohen and Presern [21] also give a self-contained treatment of prescribed traces. This appendix reconstructs the classical trace-one argument and makes its finite verification explicit; no priority is claimed for the existence result or the character-sum method. Put $q = 2^k$, $L = \mathbb{F}_{q^2}$, $K_0 = \mathbb{F}_q$, and $M = q^2 - 1 = (q - 1)(q + 1)$. Let

$$E_1 = \{x \in L : x^q + x = 1\}.$$

The trace map $L \to K_0$ is surjective and K_0-linear, so $|E_1| = q$. If $x \in E_1$, then $x \notin K_0$ and its minimal polynomial over K_0 is $X^2 + X + xx^q$. Hence it is enough to prove that E_1 contains an element of order M.

Let ψ be the nontrivial additive character of $\mathbb{F}_2$, and put $\psi_L(x) = \psi(\mathrm{Tr}_{L/\mathbb{F}_2}(x))$ and $\psi_{K_0}(a) = \psi(\mathrm{Tr}_{K_0/\mathbb{F}_2}(a))$. For a multiplicative character χ of L^*, extend χ to 0 by $\chi(0) = 0$ and define

$$g_L(\chi) = \sum_{x \in L^*} \chi(x)\psi_L(x), \qquad g_{K_0}(\chi) = \sum_{a \in K_0^*} \chi(a)\psi_{K_0}(a), \qquad S(\chi) = \sum_{x \in E_1} \chi(x).$$

If χ is nontrivial on L^*, then $|g_L(\chi)| = q$; if $\chi|_{K_0^*}$ is nontrivial, then $|g_{K_0}(\chi)| = \sqrt{q}$. Indeed, the

usual squared-modulus calculation gives

$$|g_L(\chi)|^2 = \sum_{t\in L^*} \chi(t)^{-1} \sum_{a\in L^*} \psi_L(a(1-t)) = q^2,$$

and the same argument in K_0 gives q.

We first compute $S(\chi)$. For $\alpha \in K_0$ set $E_\alpha = \{x \in L : x^q + x = \alpha\}$; these sets partition L, and $E_\alpha = \alpha E_1$ for $\alpha \neq 0$. If $\chi^{q+1} \neq 1$, then the restriction of χ to K_0^* is nontrivial. Since $E_0 = K_0$, its contribution is $\sum_{x\in E_0} \chi(x) = 0$; hence

$$g_L(\chi) = \sum_{\alpha\in K_0} \psi_{K_0}(\alpha) \sum_{x\in E_\alpha} \chi(x) = S(\chi) g_{K_0}(\chi),$$

so $|S(\chi)| = q/\sqrt{q} = \sqrt{q}$.

Suppose now that $\chi^{q+1} = 1$ and $\chi \neq 1$. Then $\chi|_{K_0^*} = 1$. Let $H = \{h \in L^* : h^{q+1} = 1\}$; since $\gcd(q-1, q+1) = 1$, the group L^* is the direct product $K_0^* \times H$. Write $x = ch$ with $c \in K_0^*$ and $h \in H$. The condition $x^q + x = 1$ becomes $c(h + h^{-1}) = 1$; thus $h \neq 1$ and $c = (h + h^{-1})^{-1}$. This gives a bijection between $H \setminus \{1\}$ and E_1, and therefore

$$S(\chi) = \sum_{h\in H,\ h\neq 1} \chi(h) = \sum_{h\in H} \chi(h) - 1 = -1,$$

because $\chi|_H$ is nontrivial.

We shall use the primitive-element indicator

$$\mathbf{1}_{\mathrm{ord}(x)=M} = \frac{\varphi(M)}{M} \sum_{\delta|M} \frac{\mu(\delta)}{\varphi(\delta)} \sum_{\mathrm{ord}(\chi)=\delta} \chi(x). \tag{S.A.1}$$

To see this, fix a generator g of L^* and write $x = g^a$. The left side is the indicator of $\gcd(a, M) = 1$. For a character with $\chi(g) = e^{2\pi i j/M}$ and order $\delta = M/\gcd(M, j)$, its unnormalized Fourier coefficient is the Ramanujan sum $c_M(j) = \mu(\delta)\varphi(M)/\varphi(\delta)$; Fourier inversion gives (S.A.1). (The displayed identity for $c_M(j)$ follows by Möbius inversion.)

Assume for contradiction that E_1 has no element of order M. Summing (S.A.1) over E_1 gives

$$0 = \frac{\varphi(M)}{M} \sum_{\delta|M} \frac{\mu(\delta)}{\varphi(\delta)} S_\delta, \qquad S_\delta = \sum_{\mathrm{ord}(\chi)=\delta} S(\chi).$$

Let $s = \omega(M)$ and $s' = \omega(q+1)$ be the numbers of distinct prime divisors. The $\delta = 1$ term is q. If $\delta \mid q+1$ and $\delta > 1$, then each character of order δ satisfies $\chi^{q+1} = 1$, so the corresponding contribution is $-\mu(\delta)$; summing over $\delta \mid q+1$, $\delta > 1$, gives 1. For the remaining squarefree divisors $\delta \nmid q+1$, the character sums are bounded by $\sqrt{q}$, and their number is at most $2^s - 2^{s'}$. Consequently,

$$q + 1 \leq (2^s - 2^{s'})\sqrt{q}. \tag{S.A.2}$$

Since $q + 1 > \sqrt{q}$, this implies

$$2^s > 2^{k/2}, \qquad \text{so} \qquad s > k/2.$$

In particular $k < 2s$. Let $p_1 < p_2 < \cdots$ be the primes and $P_s = p_1 \cdots p_s$. Since $M = 2^{2k} - 1$ has exactly s distinct prime divisors, $M \geq P_s$. We have

$$P_{16} = 32589158477190044730 > 2^{64}.$$

For $n \geq 16$, the ratio

$$\frac{P_{n+1}/2^{4(n+1)}}{P_n/2^{4n}} = \frac{p_{n+1}}{16}$$

is greater than 1, because $p_{n+1} \geq 59$. Hence $P_s > 2^{4s}$ for every $s \geq 16$. But $k < 2s$ gives

$$M = 2^{2k} - 1 < 2^{2k} < 2^{4s},$$

so $s \geq 16$ is impossible. Thus $s \leq 15$ and $k < 2s \leq 30$, i.e. $k \leq 29$.

It remains to cover this finite range. Each row below specifies an irreducible degree-k polynomial m_k over $\mathbb{F}_2$ and an element $a_k \in K_0$ with $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$, where $K_0 = \mathbb{F}_2[x]/(m_k)$. The displayed hexadecimal integers encode their binary coefficient vectors. For each row let t be a root of $X^2 + X + a_k$ in $K_0[t]/(X^2 + X + a_k)$. The deterministic verifier checks that m_k is irreducible and that $\mathrm{Tr}_{K_0/\mathbb{F}_2}(a_k) = 1$. The trace condition implies that $X^2 + X + a_k$ has no root in K_0 and is irreducible there; the quotient is therefore a quadratic field extension. The verifier then checks that $t^{2^k} = t + 1$, and that $t^M = 1$ while $t^{M/p} \neq 1$ for every prime $p \mid M$. The last two checks show that $\mathrm{ord}(t) = M$; the third shows that $\mathrm{Tr}_{L/K_0}(t) = 1$. A verifier for the table, `verify_k1_29.py`, is included as an ancillary file. These checks are performed row by row using polynomial arithmetic in characteristic two. Its SHA-256 hash is `042CE7F74EDFF13ADE5C6285E42CB3D15491FC25C65B99BC670D256047EE8D9F`.

k	m_k	a_k
1	0x3	0x1
2	0x7	0x3
3	0xb	0x5
4	0x13	0xd
5	0x25	0x13
6	0x43	0x34
7	0x83	0x47
8	0x11b	0xc8
9	0x203	0x2b
10	0x409	0xcb
11	0x805	0x525
12	0x1009	0x39e
13	0x201b	0x1ec4
14	0x4021	0x3717
15	0x8003	0x62bf
16	0x1002b	0x1e6e
17	0x20009	0x1155b
18	0x40009	0x2ee84
19	0x80027	0xb9af
20	0x100009	0x6aee2
21	0x200005	0x15750b
22	0x400003	0x24a150
23	0x800021	0x6c92f9
24	0x100001b	0x389ec4
25	0x2000009	0x6801ad
26	0x400001b	0x2707d2e
27	0x8000027	0x6e9ab1e
28	0x10000003	0xfac7cd2
29	0x20000005	0xc69b55e

The table therefore supplies the required element for every $k \leq 29$. Combining this with the contradiction for $k \geq 30$ proves the theorem. □

S.B Extended field certificates and finite checks

Main Appendix B displays the two double-order certificates for the alternative route through the relative-trace classification. Here we retain the full range $2 \le s \le 32$ and all associated field checks. The first two tables record a single full-order element; the additional exponent table gives both linked elements full order. Together with Proposition S7.1, the double-order table supplies the retained alternative existence proof. The exact certificate files, verifiers and historical outputs are supplied with the project.

A hexadecimal integer encodes a binary polynomial with its least significant bit the constant coefficient. Each row means

$$E = \mathbb{F}_2[X]/(M_s), \qquad L = E[Z]/(Z^2 + (1 + b + w)Z + b), \qquad a = Z.$$

Thus only M_s, b, w need be specified; all field operations are exact.

s	M_s (hex)	b (hex)	w (hex)
2	13	8	1
3	43	6	3b
4	11b	2f	1
5	409	9b	2c9
6	1009	c78	8e5
7	4021	3e8a	1
8	1002b	2878	689
9	40009	303be	12d39
10	100009	a3085	1
11	400003	662cc	3205f9
12	100001b	d0842d	1
13	400001b	32f9317	1
14	10000003	15a3baa	6c6a1f9
15	40000003	317d6a85	36559252
16	10000008d	4893e13d	2e37b719
17	40000001b	dcbd299	91607c0b

s	M_s (hex)	b (hex)	w (hex)
18	1000000035	78c0283c4	1
19	4000000063	1153a3d1ef	16004f10d3
20	10000000039	1f9215dca7	1
21	40000000027	2657c0cd035	374caba3695
22	100000000021	fe622270c1c	1
23	400000000003	17c45ffbd76	29bd79b1ae4
24	1000000000002d	fed891dbe6c9	ec7a6a1920af
25	400000000001d	2907b6c341c09	1
26	10000000000009	c6d27ccd2b0df	432bf4829f702
27	4000000000007d	28047cef28c34e	1
28	100000000000095	a586528c5b1c9	1
29	400000000000063	243f16ec4e21119	1
30	1000000000000003	8c388b10c413f59	1
31	4000000000000069	12dfcd22c3b1728b	1
32	10000000000000001b	ae3111b3bb81428f	c171e75cc47c2b2e

The ancillary file `inner_critical_certificates.json` also records the full factorizations of A, B, every nonidentity residue $a^{d/p}$ for a prime $p \mid d$, and recursive Lucas primality certificates for those primes. Verification consists of the following finite identities, carried out by `verify_inner_`

`critical.py`. The additional `verify_inner_symmetry.py` checks the four-torsion identities and rational Lang-fiber multiplier at all 31 certificate points, and enumerates every primitive-norm fiber for $s = 2, \dots, 6$. It checks four-point orbits, order preservation, divisibility by four and the exact integer multiplicity distributions underlying the character symmetries. The new verification logs and runtime metadata are supplied with the ancillary files; these finite checks supplement the paper proof of Proposition S6.2. First, M_s has degree $n = 2s$, and

$$X^{2^n} \equiv X \pmod{M_s}, \qquad \gcd(M_s, X^{2^{n/r}} - X) = 1 \quad (r \mid n \text{ prime}),$$

which certify irreducibility. Second, $b^A = w^A = 1$, $b^{A/p} \neq 1$ for every prime $p \mid A$, $c = 1 + b + w \neq 0$, and $\mathrm{Tr}_{E/\mathbb{F}_2}(b/c^2) = 1$. Third, in the quadratic field one checks

$$a^d = (1+a)^d = 1, \quad a^B = b, \quad (1+a)^B = w, \quad a^{d/p} \neq 1 \quad (p \mid d \text{ prime}).$$

Primality is certified without probable-prime assumptions: the base case is 2; for an integer $p > 2$ one supplies a complete recursively certified factorization of $p - 1$ and an integer v with $v^{p-1} \equiv 1 \pmod{p}$ and $\gcd(v^{(p-1)/r} - 1, p) = 1$ for every prime $r \mid p - 1$. For any prime divisor ℓ of p, these conditions force the order of v modulo ℓ to be $p - 1$, hence $\ell \geq p$ and p is prime. The independent verifier uses different field multiplication and powering implementations from the certificate finder; it requires only the Python standard library, run without the `-O` flag.

The small-prime list used in the analytic constant is

$$5, 13, 17, 29, 37, 41, 53, 61, 73, 89, 97, 101,$$
$$109, 113, 137, 149, 157, 173, 181, 193, 197, 229, 233, 241.$$

Their product is 6045408114863853991373345088496297320256672385; the integer comparison is $256^{24} < 32^8$ times this product. The verifier checks this comparison as well.

As a separate small-field audit, multiplicative-subgroup enumeration and polynomial gcds with the Möbius sieve agree on the following counts. Here N_d^{prim} has the meaning in Theorem S6.3.

s	d	T_d	$\lvert\mathcal{R}_d\rvert$	N_d^{prim}	D_d
2	85	24	24	16	8
3	585	86	72	48	24
4	4369	240	240	224	208
5	33825	1214	960	520	240
6	266305	4104	3960	2928	1992

The data also verify the genus-one Hasse bounds on every primitive norm fiber in this small range. They confirm that the diagonal $b = w$ gives no realizing slope for $s = 3, 5$. These supplementary counts do not replace the general proof. The present all-dimensional theorem uses the classical Weil bound and a finite exact certificate, and is outside the Lean formalization.

Additional double-order certificates. The file `inner_double_order_certificates.json` uses the same M_s, b as the preceding table and replaces w by b^{e_s}, with the following exponents. The root $a = Z$ of $Z^2 + (1 + b + b^{e_s})Z + b$ and its translate $1 + a$ both have exact order d, for every $2 \leq s \leq 32$.

s	e_s	s	e_s
2	1	18	1
3	4	19	5
4	1	20	1
5	4	21	5
6	3	22	1
7	1	23	8
8	4	24	4
9	26	25	1
10	1	26	3
11	7	27	1
12	1	28	1
13	1	29	1
14	8	30	1
15	4	31	1
16	4	32	3
17	7		

The independent `verify_inner_mixed.py` checks irreducibility, the trace, both norms and both nonidentity residues $a^{d/p}, (1+a)^{d/p}$ for every prime $p \mid d$. It uses the complete factorizations and recursive Lucas certificates of `inner_critical_certificates.json`; all new residues and all norm-fiber counts for $s = 2, \ldots, 6$ are recorded in `inner_mixed_verification.json`. It also checks the corrected mixed Lang functions and trace multipliers at the 31 certificate points. These computations establish the stated finite range. Together with Supplementary Proposition S7.1 for $s \geq 33$, they also give the alternative existence route. The alternative route through the complete trace classification uses only the two exceptional rows of these field certificates. Complete primitive inner-norm support uses the separate records in main Appendix A. The new verifier `verify_inner_double_all.py` independently rechecks all 31 double-order certificates, including their recursive Lucas primality proofs, without calling a certificate finder.

Associated tail-bound and trace-one audit record

The integer tail lemma and all 81 intermediate bounds are kept in main Appendix B. The accompanying field and trace-one audit record is retained below. The source results and environment record are `inner_double_all_verification.json` and `double_all_verification_environment.json`. The reorganization includes a fresh finite-verification log at `verification/author_runtime/author_double_order_audit.log`. It also exhaustively checks the trace-one subfamily for $2 \leq s \leq 6$: its counts are respectively $8, 0, 16, 0, 24$. In particular the all-dimensional claim is for D_d, not for Δ_t. These older field certificates and the integer bounds remain valid. Appendix C gives the stronger prime-sieve range and the complete trace-one classification.

S.C Prescribed-order predecessors and quantitative comparison

Put $t = 2^s$, $s \geq 2$, $A = t + 1$, $B = t^2 + 1$, $d = AB$ and $r = t - 1$. The identities $d = (t^4 - 1)/r$ and $C_d = \ker N_{L/K}$ identify the double full-order statistic with an established r-primitive-pair count. This section compares particular published estimates at these parameters. It does not assert that failure of a sufficient criterion excludes every refinement of its proof method.

S.C.1 Adjacent exact-order pairs

In [14, Proposition 3.1], set the field size to t^4, the index to r, the common difference to 1 and the length to 2. Its $N_r(t^4, 1, 2)$ equals D_d. Write $M = \theta(d)^2(t+1)^2$. Their estimate is centered at

$$M_{\mathrm{SSL}} = \theta(d)^2 \frac{t^4}{r^2}, \qquad M_{\mathrm{SSL}} - M = \theta(d)^2 \frac{2t^2 - 1}{r^2} = O(1).$$

Thus the leading density is already represented by that estimate. Its relative error bound becomes $((rW(d))^3/(rW(d)-1))/t^2$. Theorem 1 there would require

$$t^2 \geq \frac{(rW(d))^3}{rW(d) - 1}.$$

This fails: $\gcd(A, B) = 1$, $A, B > 1$, so $W(d) \geq 4$, and the right-hand side exceeds $16r^2 > t^2$. Its stronger Theorem 5 sieve cannot repair this numerical loss. Even allowing length two in that displayed criterion, let h be the number of omitted primes, $l \mid d$ the sieve core, $\delta = 1 - 2\sum_{i=1}^{h} p_i^{-1} > 0$ and $\Delta = (2h-1)/\delta + 2$. It requires

$$t^2 \geq \Delta r^2 W(l)^2.$$

If $h = 0$, then $\Delta = 1$ and $W(l) = W(d) \geq 4$. If $h \geq 1$, then $\Delta \geq 3$ and $W(l) \geq 1$. In either case its right-hand side exceeds t^2 for $t \geq 4$. This checks every legal split of that criterion, independently of the length restriction preceding the sieve section. The fixed-r asymptotic corollary does not by itself give a uniform conclusion for $r = t - 1$.

For comparison, [10, Corollary 17], with polynomials $X, X+1$, indices r, r and freeness parameters d, d, requires $t^2 > r^2 W(d)^2$, which also fails. The general r-primitive and k-normal frameworks in [11, Theorem 3.2] and [12, Theorem 3.3] have additional conditions; even their favorable numerical specialization with trivial additive factors has right-hand side $2r^2W(d)^2 > t^2$. These comparisons concern the stated sufficient inequalities, not a new formulation of adjacent prescribed orders.

S.C.2 Prescribed norms and a change of base field

Choudhary–Sharma [15, Definition 3.1] give another exact identification. Set their $q = t$, $n = 4$, $m = 2$, $\gamma = 1$, $r_1 = r_2 = r$, and $a_1 = a_2 = 1$. Then $r_i' = r$, $R_i = d$, $R_i' = 1$, $Q_{R_i} = d$, and $N_{1,1}(d) = D_d$. Their Theorem 1 becomes $1 > 2r^2W(d)^2$. For Theorem 3 write $U = \sum_i u_i$ and $\Delta = (U-1)/\delta + 2$. If $U = 0$ then $\Delta = 1$; otherwise $\Delta \geq 2$. Its right-hand side is at least $2r^2$. For every legal split in Theorem 4, the positive denominator is at most 1 and the numerator is at least $\sum_i c_i \geq 2$; the full right-hand side is at least $4r^2$.

Changing the base field cannot rescue these inequalities. If $q_0^n = t^4$, their left-hand side for length two is

$$q_0^{n/2-2} = \frac{t^2}{q_0^2} \leq \frac{t^2}{4} < 2r^2.$$

This already lies below the lower bounds for Theorems 1, 3 and 4, even when all required norm data are admissible. Their Theorem 2 assumes odd characteristic. The application with $r = 3$ is irrelevant for $s \geq 3$. When $s = 2$, the only possible base with $n \geq 7$ is $(q_0, n) = (2, 8)$, which is listed among its possible exceptions in Theorem 5.

S.C.3 The reciprocal-trace statistic

The count Δ_t in Proposition S7.1 is exactly $N_{t^4}(B, A)$ in [18, Theorem 1.2]: order B forces $\gamma \in V$, and order A forces $\gamma + \gamma^{-1} \in U$. Substitution in that theorem gives

$$\left|\Delta_t - \frac{\theta(d)}{t-1}\right| < 2\theta(d)W(d)t^2.$$

The center is much smaller than the actual norm-concentrated population, but the large error permits this. The estimate here is

$$|\Delta_t - \theta(d)(t+1)| \le \theta(d)(4W(d)-2)\sqrt{t}.$$

It is the refined geometry of the same statistic that changes the estimate; the statistic itself is prior. The substitution $a = (1+\gamma)^{-1}$ and its Frobenius relation also occur in [18, §4]. Theorem 1.5 there, with base t^2, prescribes reciprocal-trace order $t^2 - 1$, not A; that trace has norm of order $t - 1$ to K.

S.C.4 Quartic geometry and the exact affine slice

Hachenberger's formal quartic treatment [13] gives a direct geometric predecessor. In his notation set $q = t$, with his ground, intermediate and quartic fields corresponding to our K, E, L, respectively. Section 3.1 (p. 94) gives $L^\times = K^\times \times C_d$ and identifies primitive projective points with order-d norm-one representatives. Proposition 4.1 and Remark 4.2 (p. 97) give the ovoids

$$\mathcal{O}_u = \{[uv] : v \in V\}, \qquad u \in U.$$

They are primitive exactly when $\mathrm{ord}(u) = A$; each then contains $\varphi(B)$ primitive points in all. Proposition 7.1 (p. 107) shows that every $u \neq 1$ has a trace-zero oval $\mathcal{O}_u \cap H$ with $t+1$ points, where $H = \mathbb{P}(\ker \mathrm{Tr}_{L/K})$. These statements exactly overlap the ambient geometry used here.

The remaining condition has a precise projective formulation. For a primitive point $[x] \in H$, put $c = x + x^{t^2}$. Then $c \in K^\times$: trace transitivity puts c in K, and $c = 0$ would put x in the quadratic subfield, contrary to primitivity. Define

$$\rho([x]) = \frac{N_{L/K}(x)}{(x + x^{t^2})^4} \in K^\times. \tag{S.C.1}$$

Replacing x by λx, $\lambda \in K^\times$, multiplies numerator and denominator by λ^4; thus ρ is well defined. The trace-one representative $a = x/c$ has $N_{L/K}(a) = \rho([x])$. Consequently the trace-one double full-order subfamily is exactly the primitive-point level set $\rho = 1$. Indeed $\rho = 1$ gives $a \in C_d$ of order d and $a^{t^2} = 1 + a$. Conversely, such an a gives $\gamma = (1+a)/a$ of order B and $\gamma + \gamma^{-1} = a^{-B}$ of order A, recovering Proposition S7.1.

This also locates the global primitive-normal bounds relative to the count needed here. Let M_c count order-d norm-one representatives with $a + a^{t^2} = c$, $c \in K^\times$. Characteristic two makes the trace-zero plane precisely the non-normal configuration [13, §5.2.1]. Each primitive projective point has $\varphi(t-1)$ ambient primitive representatives. With $PN_4(t)$ as in [13], therefore,

$$\sum_{c \in K^\times} M_c = \varphi(d) - \frac{PN_4(t)}{\varphi(t-1)}, \qquad M_1 = \Delta_t. \tag{S.C.2}$$

Section 6.1 and Theorem 1.2 there give $\sum_c M_c \le t\varphi(A)$. Theorem 1.3 makes this an equality when B is prime. Neither is a positive lower bound on M_1. For a norm-one representative, $\rho = c^{-4}$, so division by c preserves norm one exactly when $c = 1$.

The distinction is visible in exact small parameters. For $t = 8$ and 32, the primitive trace-zero populations in (S.C.2) are 36 and 500, respectively, while $M_1 = \Delta_t = 0$ in both cases. The full double-order populations are nevertheless $D_d = 24$ and 240, respectively. Thus the nonempty primitive-ovoid oval cannot by itself prove positivity of the fixed trace-one subfamily. The estimates of Proposition S7.1 and Theorems 4.3–4.4 are not stated in that projective treatment; the additional affine distribution and order filters require further analysis. This comparison checks its full formal text and the relevant proofs; it does not claim an exhaustive survey of all subsequent refinements of quartic geometry.

Kudo–Nogata [16, Propositions 3.6 and 4.3] treat quadratic norm root counts and the ordinary cyclotomic intersection, respectively. Setting their base to t and extension degree to 4 gives the unfiltered membership count T_d. The extra joint-generation and individual-order filters require further control. In particular, Proposition 3.1 is the elementary quadratic root-count reduction, rather than the main quantitative innovation.

The curve character bound, Lang construction, and primitive indicators used here are classical [8, 7, 10]. The proofs establish their hypotheses on the specified fibers: the constant norm-character exceptions, geometric monodromy, active boundary conductors and the exact coefficient average. Together with the all-parameter closures, these are the results on which the present contribution is based. Theorem 4.5 adds complete support at each prescribed primitive inner norm. Its extra step is the three-coordinate sieve and full arithmetic closure, using the existing curve bound and the classical sieve principle; it does not assert a new character error estimate.

S.C.5 Trace statistics in norm tori

There is related recent work on trace fibers of norm tori. Shin [29, Proposition 5.6 and Theorems 5.7–5.8] treats finite étale cubic algebras in characteristic different from $2, 3$. Every smooth trace/norm fiber is equidistributed over subgroup cosets, with error at most $3(1 - 1/m)\sqrt{q}$ for index m. The proof uses Lang character sheaves and a Picard–Kummer kernel calculation; the nodal boundary has exceptional characters. Thus square-root trace equidistribution in norm tori and its geometric character analysis are already part of the literature. Those stated cubic theorems do not apply to the present characteristic-two quartic relative-trace curves. The specific descent, four boundary valuations, and exact-order classification are the results established here, rather than a new general sheaf method.

For the prescribed-order, norm and trace framework of [30], the earlier function-field count [31] and primitive-normal geometry [32], see Main Appendix E. It gives the parameter specializations, sufficient-criterion comparisons and aggregate-to-slice identities.

S.D Verification scope and original coordinate certificates

This appendix collects the detailed arithmetic checks, data-deposit record and selective formalization scope. Main Appendix D gives the shorter proof-input map. Current project paths refer to the accompanying restructured project; paths in the archived subsection refer to the original paper directory under `provenance/original_repository/`. The field tables themselves are in Appendix S.B.

Source and version record

The accompanying project contains the exact field certificates, deterministic verifiers and current sources. The archived input and its original verification materials come from

https://github.com/63331849121/yijing-mathematical-structures.

The archived input is the directory PaperA_scalar_realizations at revision d33c51696f81492de67b8df2166d0c887666176c. The archive records file hashes and the correspondence between the original and present theorem labels. The research data and selective formalization artifacts are also available with the public v17 preprint at https://doi.org/10.5281/zenodo.23721670. That versioned record contains earlier PDFs; the accompanying package supplies the present sources and verification files.

Current arithmetic checks

The complete primitive inner-norm support theorem uses the three-coordinate sieve with primitive b, the uniform tail for $s \geq 1924$, and 1922 exact integer rows for $2 \leq s \leq 1923$. The 13 remaining parameters have 1182 witnesses whose Frobenius orbits cover all 32022 primitive norms. Main Appendix A gives the proof, coverage table and independent replay methods. The data and portable verifiers are in verification/primitive_norm_2026-11-02/. This theorem is outside the retained Lean formalization.

The new distribution proofs are in Appendix S.E. Their portable arithmetic and spectral verifiers are in verification/distribution_2026-11-02/. Fresh reports include the Kloosterman moment replay, two independent cubic reductions for $s = 3, 5, 7, 9, 11$, the exact projective spectrum certificate and the sieve coefficient audit. The fixed conductor constant in the mixed estimate is not numerically evaluated, and the full-order variance bound retains its $W(A)^2$ term. These arguments are outside the retained Lean scope.

The alternative full-family existence route combines the relative-trace classification with the off-slice double-order certificates at $s = 3, 5$. Its independent tail bound uses all 81 intermediate divisor bounds for $33 \leq s \leq 113$ and the uniform inequality for $s \geq 114$. The verifier verify_inner_double_all.py checks those integer inequalities and all 31 field certificates for $2 \leq s \leq 32$; the remaining field rows are additional checks of the shorter route. The separate trace-one proof is retained in the supplement. Appendix B states these roles, and the verification files give the exact inputs and outcomes.

The earlier K-valued relative-trace verifier and its exact output are in verification/uniform_trace/. They check all 24 prime-sieve rows, all eight trace-one polynomial counts, the two Bézout identities for emptiness, and every nonzero K-valued trace slice at $s = 2, \ldots, 6$ by independent parameterizations. They also verify 1010 full-order witnesses covering every nonexceptional nonzero K-valued slice at $s = 2, \ldots, 9$. The current finite proof for the whole E^* uses the files in verification/radical_support_2026-10-26/. It has 755 central witnesses and 731 separating witnesses; 255 additional central witnesses at the analytic case $s = 8$ are redundant checks. A separate arithmetic implementation transports the data to degree-$4s$ binary polynomial fields and verifies every witness directly. The inverse-support lemma propagates positivity from the representatives to the whole trace classes. The two exceptional supports are independently checked as well. The general reduction depends only on Theorem 5.1 and elementary cyclic-group arguments, not on the completed classification. Exhaustive small cyclic-group checks and counterexamples to dropping its hypotheses are also recorded.

The earlier 3319 exact witness families in verification/extended_trace_2026-10-26/ remain a separate check on all 349512 nonzero traces for $s = 2, \ldots, 9$. Independent full-subgroup enumeration for $s = 2, \ldots, 6$ additionally checks all 5451 nonzero E-valued populations and the stated error bounds. The phase reduction and prime sieve prove the infinite range; finite checks do not replace those arguments. Appendix C distinguishes the current support proof from the retained all-trace certificates.

The independent verifier `verify_structure.py`, in `verification/fulltext_structure_2026-10-26/`, enumerates all nonzero ambient-field elements for $s = 2, \dots, 5$. It checks the convolution on all 1356 nonzero relative traces, all 36080 norm–trace scaling cases (including trace zero), the exact means, the variance inequalities and the radical periods. Separate binary-polynomial divisions verify the exceptional reciprocal-trace supports. These are independent finite checks; the general Fourier and scaling identities are proved in Section 5.

The rational-image comparison with [33] has a further independent verifier in `verification/cstc26_fulltext_2026-10-26/verify_specialization.py`. It checks the exact inverse-trace summation on all 5451 nonzero relative traces for $s = 2, \dots, 6$, including the 20 empty slices, and evaluates the rational function on every full-order element in these norm kernels. It also checks the specialized sufficient inequalities by exact rational arithmetic.

The comparisons with [31, 32] have exact checks in `verification/older_sources_2026-10-26/verify_older_sources.py`. For $s = 2, \dots, 9$ they evaluate every squarefree-core partition for the direct prescribed-order count and its primitive lift. The all-parameter obstruction follows from the inequalities in Appendix E. The script also matches the four published binary rows $t = 4, 8, 16, 32$ of [32, Table 1] against the preceding independent norm-kernel enumeration, and separately checks the distinct binary rational-function example in [31]. These are finite consistency checks, not a new full enumeration of the quartic fields or a proof of global priority.

Element-level data deposit

A separate data deposit supplies newly regenerated element-level logs for all nonzero elements of L at $s = 2, \dots, 5$, and for all elements of C_d at $s = 6$. Each record identifies the finite-field modulus, through its parameter key, the two relative norms, the norm-fiber labels and the E-valued relative trace. Population logs include zero traces and the full declared ranges. Their producer and a separate readback checker are supplied in `data_deposit/`. These records are distinguished from the unchanged original certificates and their historical run logs; they do not enumerate the infinite parameter ranges used in the proofs.

Selective Lean record retained from v17

The nine current modules in `formalization/` have been checked together with Lean 4.31.0: 129 theorem and lemma declarations, plus 33 definitions. The original seven source modules retain their previous bytes. The ten grid and cover lemmas still import only `Init` and have no axiom dependencies. All remaining audits use only the standard foundational axioms `propext`, `Classical.choice`, and `Quot.sound`; no proof placeholder or new mathematical axiom is introduced.

The existing modules construct the multiplicative exponent equivalence, the genuine finite-field root subgroup, inverse-support closure and exact two-preimage parameter counts. The new rational transfer module constructs an actual equivalence between the admissible γ and a sets, with forward map $c/(1+\gamma)$ and inverse $(a+c)/a$. It checks the Frobenius and power identities, derives the required coprimality relations from $t = 2^s$, and proves that order $d = (t+1)(t^2+1)$ is equivalent to orders $t+1$ and t^2+1 of the recovered components. Its finite-set bijection transfers the primitive-element count to the filtered parameter count, and then to twice the filtered support count.

The named norm-and-trace module works with actual finite fields $K \subset E \subset L$, $|K| = 2^s$, and $[E : K] = [L : E] = 2$. It derives the quartic degree and the finite-field norm and trace power formulas from the library's `Algebra.norm` and `Algebra.trace`. The resulting population is therefore the original M_c, not merely a γ count. For nonzero $c \in E$ it proves the exact counting transfer, and for $h \in K^*$, $N_{E/K}(u) = 1$, it proves the shifted support formula for M_{hu}. The zero

slice $M_0 = 0$ and the equivalence between zero population and empty filtered support are also checked. These statements cover the manuscript's parameter range and add no new novelty claim.

Concrete field representations, the original certificate generator and its coordinate change, finite witnesses and exceptional polynomial checks remain outside Lean. The complete one-prime and prime-kernel branches, quotient-fiber counts, unique trace-class decomposition, geometric estimates, sieve and infinite-tail arguments, and the final all-parameter assembly also remain unformalized. In particular neither the complete quantitative Theorem 5.1 nor Theorem 5.4 is claimed formalized. All source hashes, declaration audits and dependency commits are in `formalization/interfaces_check.json`; `formalization/INTERFACE_SCOPE.txt` gives the detailed boundary.

The three original Lean files have also been rerun on the distributed sources in an isolated environment using the original Lean version and all nine pinned dependency commits. All three checks exit successfully. The mathlib archive cache was reused; a cache-free cloud build is not claimed. The historical `checked_environment.json` contains different hashes for `AGL.lean` and `CohenEvenConditional.lean`; it is preserved as history. The new results and the actual distributed hashes are recorded separately in `verification/supplement_sync_2026-10-26/original_lean_fresh_checks.json`. The old hash record is not used as proof that the current bytes were checked. No source file in the original Lean archive was modified to obtain the new checks.

Archived verification record of the original manuscript

The following record is retained from the 52-page source at the pinned revision specified in the source and version record above (this appendix). Its original run counts and formalization claims describe that source record; current scope is given above.

All numerical claims (fixed points, cycle types, group order 4032, and the characteristic polynomials of L_T, L_D, and L_K) were verified computationally. The script `verify_paperA_structure.py` reproduces the coordinate, cycle, group-order, basis-ordering, and τ-collapse data, as well as the $\mathbb{F}_{64}$ instance of Proposition S3.2 and the CRT exponent convention. It computes the characteristic polynomials as $\det(tI + L)$ by the permutation expansion in characteristic two. The separate `verify_k1_29.py` checks the finite PTE certificate. The script `verify_complete_mapping_orbits.py` enumerates all binary matrices and translations for $n \leq 4$, checks Lemmas 2.1–2.2 and the coordinate-permutation identities of Proposition S3.1, and compares the resulting counts with Proposition S3.4. For $n = 4$ it enumerates line spreads by exact cover, orients every line, checks $F(f) = f$, and tests additivity, reproducing the counts following Proposition S3.3. The additional `verify_affine_groups_and_conjugacy.py` partitions all admissible matrices for $n \leq 4$ into linear-conjugacy classes and directly closes the two-generator permutation group for every translation at each matrix representative. The $4, 16, 80$ closure cases for $n = 2, 3, 4$, respectively, cover all $8, 384, 93184$ affine pairs by linear conjugacy. These closures are compared with Theorem 2.3; transitivity, block systems, and double transitivity are also tested directly. The same script checks Theorem S3.7 using all 1792 maps and all 20160 linear changes of coordinates, and verifies the signed Burnside count. It shares basic permutation and spread routines with `verify_complete_mapping_orbits.py`; the direct closure computation does not use the proposed group formula. Its data file `affine_groups_and_conjugacy_results.json` includes every matrix representative and translation case, map representatives, stabilizer matrices, and the fixed-orientation counts. These deterministic scripts require only the Python standard library and were run with Python 3.12.14. The ancillary data include the 64 rows of T, D, K, all 720 basis orderings, the 29 PTE certificates with their factorizations, the 56 four-dimensional spreads, and a nonlinear fixed-map witness. The script `export_paperA_data.py` regenerates the coordinate and certificate tables; it reuses the two original verifiers rather than constituting an independent implementation.

The scripts, data, proof sources, reproduction instructions, and the current SHA-256 manifest accompany this manuscript as ancillary files. The single command `pythonreproduce_paperA.py` checks the manifest, runs all finite verifiers, regenerates the data, and checks their hashes again. The project repository is

`https://github.com/63331849121/yijing-mathematical-structures`.

Relative to the standard basis $e_1, \dots, e_6$, the linear part of the main map is

$$L_T = \begin{pmatrix} 0 & 0 & 1 & 0 & 0 & 0 \\ 1 & 1 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 1 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 & 0 & 0 \end{pmatrix}.$$

Formalization. `CompleteMappingTriality.lean` checks the function-level rotation, $F^2(f) = \mathrm{id} \oplus f^{-1}$, $F^3(f) = f$, and the unique-fixed-point assertion of Lemmas 2.1–2.2 for additive groups of exponent two.

The affine group-generation argument (for a field $\mathbb{K}$ with a primitive element α and $c \neq 0$, the maps T and D generate $\mathrm{AGL}(1, \mathbb{K})$, the core of Theorem S1.4 and §S2) and the algebraic lemmas used in it (the commutator is a translation; conjugation of translations by T) have been formalized and machine-checked in `AGL.lean`.

The conditional file `CohenEvenConditional.lean` formalizes the field-level Frobenius identities and the resulting order-theoretic τ-collapse under an explicit order interface; it does not formalize the paper proof of the existence theorem. The concrete cycle-type data of §S1.1–S1.2 were verified by independent computation. The existence part of Theorem S2.3 is proved in Appendix S.A; its analytic part is not formalized in Lean, and its finite range is discharged by the deterministic certificate there. Thus Cohen's theorem is not used as a proof dependency; the classical references are cited only for historical attribution. The general matrix-group structure, its action criteria, the signed Burnside formula, four-dimensional conjugacy classification, coordinate-permutation description, line-spread classification, and affine orbit-count proof are paper proofs, not Lean proofs. The inverse-realization and arithmetic-sieve criteria, the primitive-group nonrealizability theorem, the positive Jacobi-sum/sieve bound, the fixed-index realization theorem and index-three threshold, the quantitative density and growing-index results, norm descent and the quarter-power obstruction, primitive inverse classification and the generating-triangle correspondence, the local invariant-orientation criterion, and the long-cycle double-transitivity criterion are also paper proofs, not part of the Lean formalization.

All three Lean files were checked using Lean 4.31.0 and mathlib commit `fabf563a7c95a166 b8d7b6efca11c8b4dc9d911f`. The ancillary `lean/` directory contains these files, `lean-toolchain`, `lakefile.lean`, and `lake-manifest.json`, pinning the proof environment. These file checks used an existing dependency cache; a fresh dependency download and cloud CI run are not claimed. Code is supplied under the MIT license, and the manuscript and data under CC BY 4.0.

The verifier `verify_standard_spread_orbits.py` independently enumerates the 60480 projective matrices for the standard six-dimensional spread and traverses all 2^{21} orientations using elementary matrix generators and the signed Frobenius action. It checks the projective cycle table, the weight polynomial, and the 80 disjoint orientation orbits. The data `standard_spread_results.json` contain every representative, its map, full linear stabilizer order, and both its orientation-orbit size and its full $\mathrm{GL}(6, 2)$ conjugacy-class size. Only this underlying spread type is covered. The verifier

`verify_nonlinear_companions.py` checks the explicit sixteen-point tables, all three cycle types, the three-cycle word certificate, and the translated-fixed identities. It does not enumerate A_{16}: the group equality has the elementary proof in Proposition S4.2. The arbitrary-dimensional asymptotic and translated-fixed results are paper proofs, outside the supplied Lean formalization.

The verifier `verify_scalar_realization.py` compares two independent finite counting methods: binary-polynomial gcds and explicit multiplicative subgroups in finite-field coordinates. It covers every divisor of 2^n-1 for $n = 2, 4, 6, 8, 10, 12, 14$, checks the Jacobi bounds and slope-generation sieve, and verifies the degree-eight witness in Corollary S5.1. The data `scalar_realization_results.json` include the index-three counts $|\mathcal{R}_d| = 0, 0, 24, 90, 396, 1722$ for $n = 4, 6, 8, 10, 12, 14$, respectively. These computations verify finite arithmetic certificates; the infinite-family and threshold assertions use the paper proofs in main Section 6 and Supplementary Corollary S5.1.

The verifier `verify_scalar_depth.py` audits the quantitative main term and error by exact rational arithmetic in even dimensions 2 through 14. It checks norm descent by polynomial gcds for $2 \le r \le 8$, all proper divisors $s \mid r$, and all $u \mid 2^s - 1$. The quarter-power counts are checked for $s = 1, 2, 3, 4, 5$ by polynomial gcds and the divisor sieve, with independent field enumeration and subfield-location checks for $s = 1, 2, 3, 4$. For primitive subgroup orders in even dimensions 2 through 14, it checks Frobenius and six-ratio orbits, the degree of the polynomial encoding the realizing slopes, and the conjugacy-class counts. It also directly enumerates generating triangles and their affine stabilizers on 4 and 16 vertices. The corresponding data are `scalar_depth_results.json`. These are finite arithmetic audits; the uniform exponent obstruction and classification use the paper proofs.

Bases for the coordinate presentation

For the coordinate presentation $T(x) = (2^{2k} - 1) \oplus \mathrm{coords}_G(x)$ on $2k$ bits the following bases give the stated group order $\langle T, D\rangle$ and K-cycle type:

k	basis G	$\langle T, D\rangle$	K type	factor
1	$(1, 3)$	$12 = 4 \cdot 3$	$1 + 1 \cdot 3$	Φ_3
2	$(1, 2, 5, 13)$	$240 = 16 \cdot 15$	$1 + 3 \cdot 5$	Φ_5
3	$(63, 33, 18, 49, 1, 21)$	$4032 = 64 \cdot 63$	$1 + 7 \cdot 9$	Φ_9
4	$(134, 136, 241, 71, 34, 65, 3, 143)$	$65280 = 256 \cdot 255$	$1 + 15 \cdot 17$	a degree-8 factor of Φ_{17}

All four rows were verified by independent computation. The 720-case count was reproduced by enumerating all orderings of the six displayed vectors, solving the corresponding $\mathbb{F}_2$ linear system for coords_G, and recording the cycle types of T, D, K whenever the two inverses exist.

Explicit tau-collapse data

With $\mathbb{F}_{2^{2k}} = \mathbb{F}_2[t]/(f)$ and elements written as polynomials in t, the following rows realize the order conditions of Theorem S2.3:

k	f	γ	$\tau = \gamma \oplus \gamma^{-1}$	$\alpha = (1 \oplus \gamma)^{-1}$
1	$t^2 + t + 1$	t	1	t
2	$t^4 + t + 1$	t^3	$t^2 + t + 1$	t
3	$t^6 + t + 1$	$t^2 + t$	$t^4 + t^3 + 1$	$t^5 + t^3 + t^2$
4	$t^8 + t^4 + t^3 + t^2 + 1$	$t^3 + t^2 + t + 1$	$t^7 + t^4 + t^3 + 1$	$t^6 + t^4 + t^3 + t^2 + 1$

In each row $\mathrm{ord}(\gamma) = \mathrm{ord}(1 \oplus \alpha^{-1}) = 2^k + 1$, τ is primitive in $\mathbb{F}_{2^k}$, and $\mathrm{ord}(\alpha) = \mathrm{ord}(1 \oplus \alpha) = 2^{2k} - 1$ (all verified). The defining polynomial f is a field-representation choice and need not coincide with

the characteristic polynomial p_T of (S1.1); for instance, the $k = 3$ row uses $f = t^6 + t + 1$, while $p_T(t) = t^6 + t^5 + t^3 + t^2 + 1$.

S.E Primitive inner-norm moments and the cubic sieve

This appendix proves Theorems 4.7 and 4.8. Its finite spectral certificate is an algebraic input to a general representation-theoretic argument. The moments use the uniform measure on the generators $\mathcal{P}$ of U, not on E^* or on isomorphism classes. The basic character indicators, Dickson identity, Artin–Schreier ramification calculation and cohomological bounds used below are classical.

S.E.1 The exact unsieved Kloosterman parameter

Write $\psi_K(x) = (-1)^{\mathrm{Tr}_{K/\mathbb{F}_2}(x)}$ and similarly for E. For $b^3 \neq 1$, take $u^2 = b$ in U and put $h = u + u^{-1} \in K \setminus \{0, 1\}$. Write $a = uv$ with $v \in V$, $\tau = v + v^{-1}$ and $z = h + \tau$. The second inner norm is $w = uz$, so its membership in U is equivalent to $z \in U$. As $K^* \cap U = \{1\}$ and $h \neq 1$, every $h + z$ is nonzero. The polynomial $v^2 + \tau v + 1$ has two roots in V exactly when $\mathrm{Tr}_{E/\mathbb{F}_2}(1/\tau^2) = 1$, and none otherwise. Thus

$$N_b = A - S(h), \qquad S(h) = \sum_{z \in U} \psi_E\big(1/(h + z)\big).$$

There is no omitted zero-trace-coefficient branch: $1 + b \in U$ would force $h = 1$.

For $x = z + z^{-1} \in K$,

$$\mathrm{Tr}_{E/K} \frac{1}{h + z} = \frac{x}{h^2 + hx + 1}.$$

The point $z = 1$ contributes one. The other x have two preimages exactly when $\mathrm{Tr}_{K/\mathbb{F}_2}(1/x) = 1$. With $H = h^2 + 1$, substitution $y = 1/x$ gives

$$S(h) = 1 + 2 \sum_{\mathrm{Tr}_K(y) = 1} \psi_K\big(1/(Hy + h)\big).$$

Now $\rho = h/H = r + r^2$ for $r = 1/(h + 1)$, so $\mathrm{Tr}_K \rho = 0$. Translate $Y = y + \rho$ and set $\lambda = 1/H$. The trace-one condition is preserved, and

$$\begin{aligned} S(h) &= 1 + \sum_{Y \neq 0} \psi_K(\lambda/Y) - \sum_{Y \neq 0} \psi_K(Y + \lambda/Y) \\ &= -\,\mathrm{Kl}_t(\lambda). \end{aligned}$$

Finally $h^2 = b + b^{-1}$, so $\lambda = \beta_b$, proving the exact identity in Theorem 4.7.

S.E.2 The five-point model and the two unsieved moments

Use a fixed characteristic-zero $\overline{\mathbb{Q}}_\ell$ coefficient field, with $\ell \neq 2$. The unnormalized Kloosterman sheaf Kl_2 has trace $-\,\mathrm{Kl}_t$, rank two, weight one, determinant $\mathbb{Q}_\ell(-1)$, tame monodromy at zero and Swan conductor one at infinity. Katz's normalized sheaf has weight zero; undoing its constant half-Tate twist gives these normalizations. Its geometric group is $\mathrm{Sp}_2 = \mathrm{SL}_2$, also in characteristic two [34, §11.0.2, Theorem 11.1].

Let $\mathcal{U}/K$ be the nonsplit norm-one torus and C its smooth compactification. In a geometric split coordinate b,

$$\beta(b) = \frac{b}{b^2 + b + 1}, \quad \beta - 1 = \frac{(b + 1)^2}{b^2 + b + 1}, \quad \beta' = \frac{(b + 1)^2}{(b^2 + b + 1)^2}.$$

Inversion invariance makes β descend to K. Delete $0, \infty, 1$ and the two nontrivial cube roots of unity to obtain Y/K. These are five geometric points; in particular the descent interchanges 0 and ∞. The map $Y \to \mathbb{P}^1 \setminus \{0, 1, \infty\}$ is finite étale of degree two. It is separable even in characteristic two. Its only ramified point before deletion is $b = 1$. Finite-index restriction preserves the connected group SL_2.

For any character χ of U, the Lang sheaf $\mathcal{L}_\chi$ is finite order and pure of weight zero. Geometrically it is Kummer type, with possible ramification only at $0, \infty$, and that ramification is tame since A is odd. Put $\mathcal{F} = \beta^* \mathrm{Kl}_2$. The standard SL_2 representation and its characteristic-zero symmetric square have no trivial subobject or quotient, even after tensoring with $\mathcal{L}_\chi$: restriction to the finite-index kernel of χ retains the same connected geometric group. Thus H^2_c vanishes by its description through geometric coinvariants; H^0_c vanishes because Y is nonproper.

For $\mathcal{F} \otimes \mathcal{L}_\chi$, the two simple poles of β have Swan conductor one each. All other boundary Swan conductors are zero. The Euler–Poincaré formula gives $\dim H^1_c = 2(5-2) + 2 = 8$. The weight bound and trace formula give

$$\left| \sum_{b \in Y(K)} \chi(b) \, \mathrm{Kl}_t(\beta_b) \right| \leq 8t.$$

For $\mathrm{Sym}^2 \mathcal{F}$, the maximum break at either pole is at most one: the original Swan conductor one bounds the maximum break by one, and tensor products do not increase that maximum. Rank three therefore gives Swan at most three at each pole. This conservative argument needs no refined local formula in characteristic two. Hence $\dim H^1_c \leq 3(5-2) + 6 = 15$. Since the determinant is $\mathbb{Q}_\ell(-1)$, the pointwise trace is $\mathrm{Kl}_t(\beta_b)^2 - t$, and

$$\left| \sum_{b \in Y(K)} \chi(b)(\mathrm{Kl}_t(\beta_b)^2 - t) \right| \leq 15t^{3/2}.$$

The cohomological formulas and weights are the classical ones in [34, §2.3, §3.6]. The deliberately deleted lisse point $b = 1$ is included in the five-point Euler characteristic.

All primitive b lie in $Y(K)$. Their indicator has absolute coefficient mass $\theta(A)W(A)$, whereas $\#\mathcal{P} = \theta(A)A$. Divide the two bounds by this size. Writing $m = \mathbb{E}k_b$ and $M_2 = \mathbb{E}k_b^2$, use $\mathrm{Var}(N_b) = M_2 - m^2$ to obtain the displayed bounds. Finally $W(A) = t^{o(1)}$ gives the stated asymptotic.

S.E.3 The cubic coordinate as a genus-two trace

Assume now that $s \geq 3$ is odd. For a nontrivial cubic character η of U, put

$$S_\eta(h) = \sum_{z \in U} \eta(z) \psi_E(1/(h+z)).$$

The preceding parametrization and $\sum_U \eta = 0$ give

$$\sum_w \eta(w) H(b, w) = -\eta(u) S_\eta(h).$$

Inversion of z shows $S_\eta = S_{\eta^{-1}}$; this real element of $\mathbb{Z}[\zeta_3]$ is an integer. Since $\eta(u) = \eta(b)^2$ and primitive b is not a cube, $\eta(u) + \eta(u)^{-1} = -1$. Therefore $R_{3,b} = S_\eta(h)$. For good cube b the sum is instead $R_{3,b} = -2S_\eta(h)$.

Write $D_3(X) = X^3 + X$ and $n_x = \#\{y \in K : D_3(y) = x\}$, counting distinct roots. For nonzero x, take $z + z^{-1} = x$. If $\mathrm{Tr}_K(1/x) = 0$, then $z \in K^*$; as $\gcd(3, t-1) = 1$, the unique cubic root in K^*

gives $n_x = 1$. The two other formal roots are interchanged by Frobenius and do not belong to K. If $\mathrm{Tr}_K(1/x) = 1$, then $z \in U \setminus \{1\}$, and the identity $D_3(r + r^{-1}) = r^3 + r^{-3}$ gives three K-roots when $z \in U^3$ and none otherwise. Thus in that case $n_x = 1 + \eta(z) + \eta(z)^{-1}$. At zero, $n_0 = 2$ because $D_3(y) = y(y+1)^2$; at infinity there is one distinct preimage.

Set $H_0 = h^2 + 1$ and $f_h(x) = x/(H_0 + hx)$. This Möbius transformation has the unique pole $x_0 = h + h^{-1}$, a split value since the associated roots are $h, h^{-1} \in K^*$. Hence $n_{x_0} = 1$. Also $f_h(0) = 0$ and $f_h(\infty) = 1/h$. The full finite-value character sum of this Möbius map on $\mathbb{P}^1(K)$ is zero. Apply the above different-root counts to its pullback by D_3. The point zero contributes the extra one; the point infinity and the deleted pole have no change of multiplicity. Therefore

$$\sum_{\substack{y \in \mathbb{P}^1(K) \\ f_h(D_3(y)) \text{ finite}}} \psi_K(f_h(D_3(y))) = 0 + 1 + (S_\eta(h) - 1) = S_\eta(h).$$

The relative-trace identity used here is $\mathrm{Tr}_{E/K}(1/(h+z)) = f_h(z + z^{-1})$.

The denominator $H_0 + hD_3(X)$ has three distinct geometric roots: its derivative is $h(X+1)^2$, and $X = 1$ is not a root. The numerator does not vanish at any such root. Infinity is regular. Each simple Artin–Schreier pole has different exponent two, as follows locally from the uniformizer $1/Y$ and the automorphism $Y \mapsto Y + 1$. The cover is geometrically connected, and Riemann–Hurwitz gives $2g - 2 = -4 + 6 = 2$. A rational simple pole contributes exactly one point on the smooth model. There is one such rational pole and a conjugate pair of other poles. The character-sum identity consequently gives $\#C_h(K) = t + 1 + S_\eta(h)$, so $R_{3,b} = -a_h$. The genus-two Weil bound proves $|R_{3,b}| \le 4\sqrt{t}$.

S.E.4 A projective specialization criterion

The following elementary representation criterion lets one exclude a common geometric component using an exact spectrum. The condition concerns all finite powers, so the certificate must exclude roots of unity, not just equality at the base point.

Lemma S.E.1 (Projective specialization)**.** *Let $X/\mathbb{F}_p$ be a smooth geometrically connected curve, and let $\mathcal{F}, \mathcal{G}$ be lisse characteristic-zero ℓ-adic sheaves, $\ell \neq p$. Suppose $\mathcal{F}$ has rank two and is geometrically absolutely irreducible. At a point $x \in X(\mathbb{F}_{p^d})$ let their nonzero, algebraic Frobenius eigenvalues be β_1, β_2 and $\alpha_1, \dots, \alpha_r$. If none of*

$$\frac{\alpha_i/\alpha_j}{\beta_1/\beta_2}, \qquad \frac{\alpha_i/\alpha_j}{\beta_2/\beta_1} \quad (1 \le i, j \le r)$$

is a root of unity, then every lisse rank-one sheaf $\mathcal{L}$ defined over a finite constant-field extension satisfies

$$\mathrm{Hom}_{\pi_1(X_{\overline{\mathbb{F}}_p})}(\mathcal{G} \otimes \mathcal{L}, \mathcal{F}^\vee) = 0.$$

Proof. Otherwise the geometric space $H = \mathrm{Hom}(\mathcal{F}, \mathcal{G}^\vee \otimes \mathcal{L}^\vee)$ is nonzero. It is stable under arithmetic conjugation, with trivial geometric action, and is therefore a constant-field representation. Absolute irreducibility of $\mathcal{F}$ makes the evaluation map $\mathcal{F} \otimes H \to \mathcal{G}^\vee \otimes \mathcal{L}^\vee$ injective. Indeed a nonzero kernel in the direct sum of copies of the simple geometric representation $\mathcal{F}$ would contain a copy defined by a nonzero element of H, contradicting its definition as an actual homomorphism. No semisimplicity of $\mathcal{G}$ is required.

Choose a finite field defining all sheaves and making x rational, with degree e over $\mathbb{F}_{p^d}$. If γ is any Frobenius eigenvalue on H, the injection places $\beta_1^e\gamma$ and $\beta_2^e\gamma$ among $\alpha_i^{-e}\delta^{-1}$, where δ is the rank-one point value. Their ratio gives $(\beta_1/\beta_2)^e = (\alpha_j/\alpha_i)^e$ for some i, j, contradicting the hypothesis. Neither γ nor δ needs to be a root of unity. □

For the application use the fixed torus $\mathcal{T} = \ker N_{\mathbb{F}_4/\mathbb{F}_2}$ over $\mathbb{F}_2$. In its geometric coordinate u, delete $0, \infty, 1, \zeta_3, \zeta_3^{-1}$ to obtain Y. The function $h = u + u^{-1}$ descends to $\mathbb{F}_2$. For odd s, $\mathcal{T}(\mathbb{F}_t) = U$. Put

$$\mathcal{F} = (u \mapsto (1+h^2)^{-1})^* \operatorname{Kl}_2, \qquad \mathcal{G} = R^1\pi_*\mathbb{Q}_\ell,$$

where π is the genus-two family (4.9). The pullback defining $\mathcal{F}$ factors through the preceding separable degree-two β and the purely inseparable map $b = u^2$. The latter is a universal homeomorphism and preserves the étale fundamental group [35, Tag 0BQN]; hence $\mathcal{F}$ retains SL_2 geometric monodromy.

For completeness the family has a smooth proper model on all of Y. Write $D = X^3 + X$, $Q = h^2 + 1 + hD$ and set $V = QY$. Its finite chart is $V^2 + QV = DQ$. At $Q = 0$, $V = 0$ and the derivative in X is $DQ' \neq 0$; elsewhere the derivative in V is $Q \neq 0$. At infinity set $x = 1/X$, $v = x^3V$, $d = 1 + x^2$, $q = h(1+x^2) + (h^2+1)x^3$. The equation is $v^2 + qv = dq$, with $q = h \neq 0$ at $x = 0$. These charts glue to a finite double cover of $\mathbb{P}^1_Y$, smooth over Y and proper. Thus $\mathcal{G}$ is lisse of rank four and pure of weight one.

S.E.5 The exact spectral certificate and the mixed moment

Choose $h \in \mathbb{F}_8$ satisfying $h^3 + h + 1 = 0$ and take $u^2 + hu + 1 = 0$ in $\mathbb{F}_{64}$. Since $\mathrm{Tr}_{\mathbb{F}_8/\mathbb{F}_2}(1/h) = 1$, the quadratic is irreducible. Its norm-one roots have neither order one nor order three, so u has order nine, so this is a good point of $Y(\mathbb{F}_8)$, and $(1+h^2)^{-1} = h$. Direct exact counts give

$$\operatorname{Kl}_8(h) = -1, \quad \operatorname{Kl}_{64}(h) = 15, \quad \#C_h(\mathbb{F}_8) = 13, \quad \#C_h(\mathbb{F}_{64}) = 75.$$

The first two numbers give trace one and determinant eight for $\mathcal{F}$. The curve counts give the first two power traces $-4, -10$ for $\mathcal{G}$. Its polarization gives the reciprocal characteristic polynomial, so

$$P_F(T) = T^2 - T + 8, \qquad P_G(T) = T^4 + 4T^3 + 13T^2 + 32T + 64.$$

The quadratic-extension counts determine Frobenius squared on this same fiber; they do not apply the cubic sieve to even s.

Set $R_F(Y) = 8Y^2 + 15Y + 8$, and define

$$R_G(Z) = \operatorname{Res}_T(P_G(T), P_G(ZT)), \qquad R = \operatorname{prim}\operatorname{Res}_Y(R_F(Y), R_G(ZY)).$$

The roots of R are exactly the spectral quotients in Lemma S.E.1. Its primitive part has degree 32 and the exact factorization $R = Q_2^4 Q_8 (Q_8')^2$, where

$$\begin{aligned}
Q_2 &= 8Z^2 + 15Z + 8,\\
Q_8 &= 4096Z^8 - 9600Z^7 + 10377Z^6 - 11100Z^5 + 12598Z^4\\
&\qquad - 11100Z^3 + 10377Z^2 - 9600Z + 4096,\\
Q_8' &= 4096Z^8 - 2880Z^7 + 5232Z^6 - 3645Z^5 + 4394Z^4\\
&\qquad - 3645Z^3 + 5232Z^2 - 2880Z + 4096.
\end{aligned}$$

No irreducibility assertion about these factors is needed. If an Nth primitive root of unity is a root of a rational factor of degree at most eight, then $\varphi(N) \leq 8$. For every N, $\varphi(N)^2 \geq N/2$: multiply $p^{a-2}(p-1)^2$ over $p^a \| N$; only 2^1 can contribute less than one, namely $1/2$. Thus $N \leq 128$. The complete list with $\varphi(N) \leq 8$ is

$$1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12, 14, 15, 16, 18, 20, 24, 30.$$

Exact polynomial Euclid gives $\gcd(\Phi_N, Q_2) = \gcd(\Phi_N, Q_8) = \gcd(\Phi_N, Q_8') = 1$ at every listed order. The distributed verifier independently reconstructs both resultants using integer Sylvester determinants and exact interpolation, and checks all 54 gcds. This excludes every possible root of unity in R.

Lemma S.E.1 therefore gives, for every relevant Lang sheaf,

$$\mathrm{Hom}_{\pi_1(Y_{\overline{\mathbb{F}}_2})}(\mathcal{G} \otimes \mathcal{L}_\lambda, \mathcal{F}^\vee) = 0.$$

Equivalently $\mathcal{F} \otimes \mathcal{G} \otimes \mathcal{L}_\lambda$ has no geometric trivial quotient, so $H_c^2 = 0$. Also $H_c^0 = 0$. For a fixed $\ell \neq 2$ define

$$C_0 = 24 + \sum_{x \in C_{\overline{\mathbb{F}}_2} \setminus Y} \mathrm{Swan}_x(\mathcal{F} \otimes \mathcal{G}). \tag{S.E.1}$$

This is a finite integer attached to a fixed family over $\mathbb{F}_2$. All relevant Lang twists are tame at $0, \infty$ and lisse at the other boundary points, so they do not change the Swan sum. The rank-eight five-point Euler characteristic gives $\dim H_c^1 = C_0$, uniformly in s and the order of λ. Weight two of the tensor product gives the character-sum bound $C_0 t^{3/2}$.

Finally square u to b. This permutes the primitive elements. At these points the tensor trace is $(-k_b)a_h = k_b R_{3,b}$. At good cube points the trace interpretation differs, but their primitive indicator is zero. Expand that indicator, whose absolute coefficient mass is $\theta(A)W(A)$, and divide by $\varphi(A) = \theta(A)A$. This proves (4.10). Centering costs at most $|\mathbb{E}k_b|\,|\mathbb{E}R_{3,b}| \le 32W(A)t^{3/2}/A$, which proves the asserted covariance bound.

Reproduction and scope. The fresh programs in `verification/distribution_2026-11-02/` check the complete finite counts, the cubic trace identity, the spectral certificate and the exact primitive-indicator coefficients. Their finite ranges and mathematical dependencies are recorded in `REPRODUCE.txt`. The general exclusion of Lang-twisted common components uses the proved specialization criterion; no complete classification of the rank-four monodromy group, full-order variance main term or numerical evaluation of C_0 is claimed. The historical Lean records do not formalize these new arguments.

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